

Auditing Knowledge-Graph-Augmented Legal Question Answering: Conformance, Baseline Contamination, and Benchmark Headroom

M. Mohaiminul Islam and Muhammad Abulaish

Abstract—Reports that a knowledge graph (KG) improves question answering (QA) usually rest on one comparison against one baseline. This paper asks what such a comparison can establish, and audits its own KG-augmented legal QA system until the answer is known. IPROn is a KG for Indian intellectual property statutes, linked to every question and answer in an extractive benchmark. Three audits structure the paper — graph conformance, evaluation integrity, and benchmark headroom — and the third is extended with a passage-withheld retrieval experiment. First, a conformance test asks whether a graph instantiates the ontology published with it: only 19.28% of the edges that assert legal facts satisfy their declared argument types, and the residual survives schema completion, endpoint typing and argument reversal. A sensitivity analysis separates the extractor’s contribution from the ontology’s, and finds that two thirds of the failures involve an endpoint the pipeline never assigned a specific class. Second, the system’s own baseline is shown to carry the graph: retrieved facts are prepended to the question in every condition, including the one labelled reader-only. Crossing the two routes by which the graph reaches the answer gives 84.88 F1 with no graph input, 83.39 with the question prefix, and 84.45 with re-ranking alone. Both routes lower F1 in isolation, though neither interval excludes zero, and reading the reported 0.00 F1 for re-ranking as the graph’s total contribution is what the contaminated baseline makes misleading. Third, the benchmark itself is measured: the gold answer lies inside the supplied passage in 100% of records and the question names its own statutory section in 94.5%, so the benchmark leaves no headroom for supplying a missing fact and very little for identifying an unnamed provision, while span-selection effects remain: re-ranking is worth +1.75 F1 to a weak reader with an interval excluding zero, whereas the high-versus-low conformance contrast is -0.07 F1 with an interval spanning zero. The passage is then withheld and retrieval required over all 1,116 statutory sections. IPROn ranks the correct section first for 1 of 344 test questions and shows no detectable contribution to a text-based hybrid retriever, and when the reader must use retrieved evidence rather than the gold passage, F1 falls from 84.88 to 43.86. A stronger section-blinded condition cannot be constructed from this benchmark, because the section reference is the entire question in 58.72% of test records. The QA findings are therefore scoped to these task formats; the instruments are not: the validator code runs without modification on two legal KGs published by another group, after the minimal syntax repair their files need in order to parse at all, and finds defects in both. The validator, audit scripts and pipeline are provided with this submission and will be archived publicly on publication.

Index Terms—Legal NLP, knowledge graph, ontology conformance, Indian IPR law, question answering, negative results,

evaluation methodology, InLegalBERT.

I. INTRODUCTION

INDIAN intellectual property rights law is written across seven separate Acts. The Patents Act, Trade Marks Act, Copyright Act, Designs Act, Geographical Indications of Goods Act, Semiconductor Integrated Circuits Layout-Design Act, and the Protection of Plant Varieties and Farmers’ Rights Act each have their own procedures, authorities, terms of protection, penalties, and remedies. They also share structure: an application is filed before a registrar or controller, a right is granted to a holder, an infringement leads to a remedy or a penalty, and one section refers to another. This mix of shared structure and Act-specific wording makes Indian IPR law a natural target for a knowledge graph.

A knowledge graph stores facts as typed links between typed things, for example patent $\rightarrow$ grantedBy $\rightarrow$ controller. Such a graph offers two things to a legal QA system. It can give a reading model extra evidence, and it can show why an answer was chosen. The second point matters in law, because a user needs to check the support for an answer, not just read it.

This work builds IPROn: an ontology for Indian IPR statutes, a graph extracted from statutory text, links from every benchmark question and answer into that graph, graph embeddings, and a QA system that uses graph evidence when it picks an answer. The first results looked good. They did not survive checking, and what the checking found is the subject of this paper.

The paper is therefore an audit rather than a system description, and it audits three things in turn: the graph against the ontology published with it, the evaluation against the baseline it claims to beat, and the benchmark against the premise that external knowledge is needed at all. Each audit was prompted by a specific failure in the first version of this work, and each generalises past it. The third is the one that decides how far the QA conclusion travels, so it is stated at the outset rather than deferred to the limitations: in this benchmark the gold answer lies inside the passage the reader is given in 100% of records, and the question names its own statutory section in 94.5%. The benefits that usually motivate a legal KG therefore have almost nowhere to appear: there is no headroom at all for supplying a fact the passage omits, and very little for finding a provision the question does not name. Effects that work through span

selection remain available and are measured in Section VI-C. No result reported here should be read as evidence about tasks of a different shape.

Figure 1 shows the whole system and where the conformance validator sits. The validator is shown beside the pipeline rather than inside it, because it is applied to a finished graph and does not depend on how that graph was made.

The graph must be checked before it is used as QA evidence. An ontology defines which types of entity a relation may join: `grantedBy` requires an intellectual property right as its subject and an authority as its object. A validator checks every edge against these rules, and only 305 of the 1,582 edges that state legal facts (19.28%) satisfy the declared types. Adding missing types and relations, or reversing the argument order, does not solve the problem. Because a low score could equally mean the ontology declared its argument types too narrowly, a sensitivity analysis bounds that possibility: refitting every domain and range to the most common class actually observed lifts conformance to 79.96%, but largely by declaring the constraint away, and two thirds of the failures involve an endpoint the pipeline never assigned a specific class at all.

The initial QA gain was also misleading, and in two separate ways. The comparison reader had been trained from a plain encoder because a checkpoint path did not resolve. After restoring the intended reader and setting a suitable decoder length, the same graph pipeline added 0.00 F1 against that baseline. Measured that way, re-ranking is worth +1.07 F1 to the weakest of ten distinct readers and +0.02 F1 to the strongest, falling as reader strength rises ($r = -0.865$, 95% CI $[-0.99, -0.06]$).

That baseline, however, was itself receiving graph evidence: the pipeline prepends retrieved facts to the question before the reader sees it, in every condition including the one labelled reader-only. Crossing the two routes separates them. A reader given no graph input at all reaches 84.88 F1; the question prefix alone costs 1.49 F1 and re-ranking alone costs 0.43 F1. Both routes lower F1, and the 0.00 is what results from measuring one of them on top of the other. A gain against a weak baseline is not by itself evidence that a KG improves QA in general, and a gain against a baseline that already receives graph evidence does not measure the graph’s total contribution.

The low conformance is not inherent to legal extraction. With the corpus, ontology, and measure fixed, rule patterns reach 21.90% conformance, the OpenIE pipeline reaches 5.91%, and the combined pipeline reaches 5.33%. A local LLM using the same ontology reaches 42.62% to 44.56% and produces more facts per sentence. On the full corpus, it produces 1,870 triples, more than the pipeline’s 1,332, at 36.10% conformance. This identifies relation assignment as a major bottleneck, alongside incomplete specific entity typing, and shows that extraction conformance can be improved substantially, though 36.10% still leaves most triples non-conformant.

Repairing it, however, does not repair the answers. A graph built by enforcing the declared types at construction time is 100% conformant and performs worse than an equally sized graph drawn at random from the unenforced layer, because enforcement discards four fifths of the edges and the lost

coverage costs more than the gained schema conformance.

Two of these findings are about instruments and one is about a graph, and they travel differently. The conformance validator and the baseline check apply to any ontology-guided KG and any KG-augmented reader; the validator is demonstrated here on two graphs built by other authors in another jurisdiction. The QA result applies to extractive benchmarks whose answers are already in the passage. Keeping those apart is what the rest of this paper tries to do.

The methodological contribution is a reproducible audit workflow for determining whether an apparent KG gain can be attributed to valid graph evidence, rather than a new ontology-checking algorithm or QA architecture. The main contributions of this paper are as follows.

- 1) A conformance audit framework for ontology-guided KGs, rather than a new reasoning algorithm. Domain and range checking under subsumption is standard; what is contributed is its packaging as a layered instrument, separating the edges that assert legal facts from those that only record where a phrase occurs, decomposing failures into the three checks C1–C3, bounding how much of the residual is attributable to the ontology’s own declarations rather than the extractor, and carrying the same measure through to the evidence the downstream re-ranker actually consumes. The instrument is shown to transfer: the same validator code runs on two legal KGs published by another group, after the minimal syntax repair their files need in order to parse, and finds defects in both.
- 2) A demonstration that a KG-augmented reader can be contaminated by its own baseline, and a procedure for detecting it. The graph reaches the answer by two routes here, only one of which the ablation switches governed, so every reported comparison remained internally valid while pointing the wrong way about the graph. The check is mechanical and is stated in Section XIII.
- 3) A measurement of what the benchmark can support, offered as a precondition for negative results of this kind. The headroom figures given above narrow what any result on this benchmark can mean, and make the mechanisms that do remain few enough to identify individually. The measurement is then pushed one step further by withholding the supplied passage and requiring retrieval over all 1,116 statutory sections. That experiment separates a limitation of the benchmark from a failure of the graph, quantifies how far reported performance changes when the supplied passage is replaced by retrieved evidence, and shows that a section-blinded condition cannot be constructed reliably from the existing questions. The figures are given in Section XII.
- 4) A controlled test of whether ontology conformance changes downstream QA, with size, scorer and reader headroom held fixed. Three graphs spanning the achievable range of conformance, up to a type-enforced 100%, are contrasted in a setting where the presence of a graph is worth a measurable amount. The conformance contrast is not distinguishable from zero and hard type filtering is actively harmful (Section VI-C). Conformance is the property the contrast is built around rather than the only one that differs,

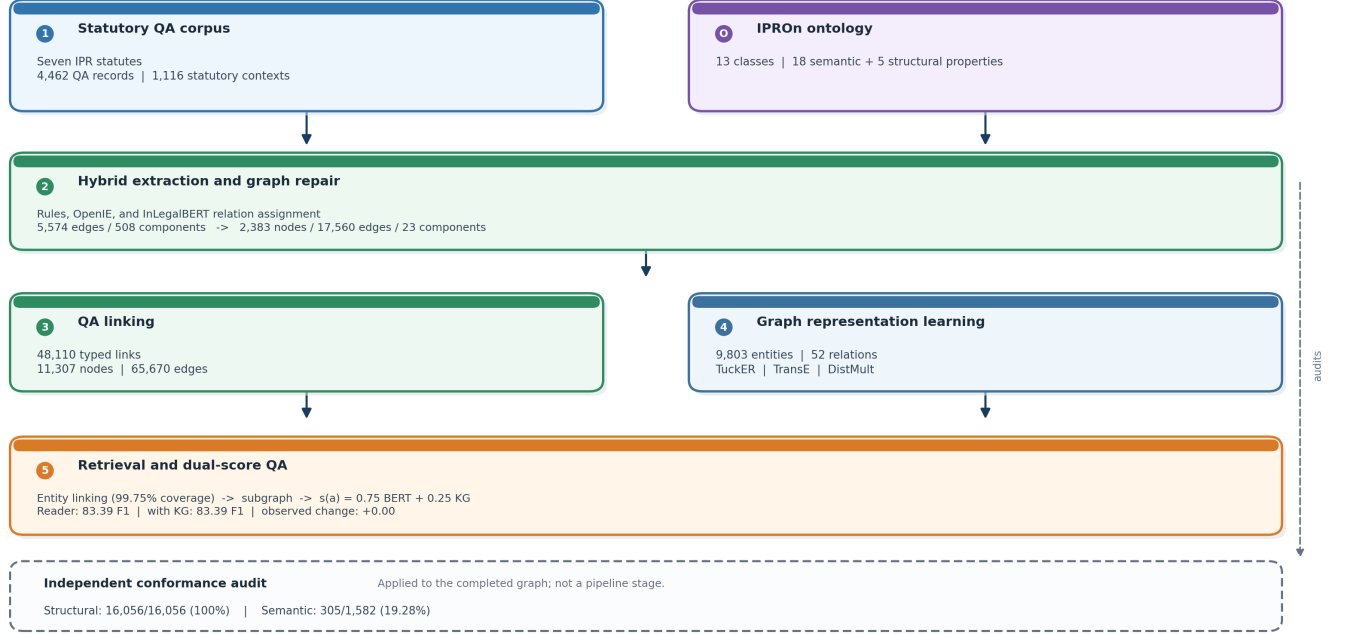


Fig. 1. IPROn: five pipeline phases and the conformance instrument. Counts are read from the released artifacts. The graph is large and well connected, and its semantic layer is 19.28% conformant. The +0.00 F1 shown in stage 5 is the comparison this system originally reported, between two conditions that both receive graph input; Section XI-A shows that a reader given no graph input at all reaches 84.88 F1, and that both routes lower F1, although neither decrease is statistically distinguishable from zero.

and the identification limits are stated with the result. IPROn itself, with 2,383 legal nodes, 17,560 legal edges and 48,110 typed links from all 4,462 benchmark questions and answers, is released alongside the validator, the audit scripts, and the JSON reports behind every table.

The rest of this paper is organised as follows. Section II reviews related work on legal QA, legal ontologies, and knowledge graphs for QA. Section III describes the dataset, the task, and how much room the task leaves for external knowledge, and Section IV describes the ontology and how the graph is built. Section V defines the conformance measure and reports what it finds, and Section VI asks whether a different construction method does better and whether a more conformant graph changes the answers. Sections VII to X cover the QA links, the retrieval procedure, the graph embeddings, and the QA model. Section XI reports the QA results and Section XII reports what happens when the supporting passage is withheld and the provision has to be retrieved, Section XIII discusses them, and Sections XIV to XV give the threats, limitations, and future work.

II. BACKGROUND AND RELATED WORK

This paper brings together three areas: extractive QA over legal text, legal ontologies and knowledge graphs, and knowledge graphs for QA. This section notes what each area usually reports and what it often leaves unchecked.

A. Extractive legal QA

Extractive QA asks a model to select an answer span from a passage. SQuAD established this task at scale [1].

SQuAD 2.0 added unanswerable questions [2], and BERT made span extraction a standard fine-tuning task [3]. Legal text is more difficult because passages are long, clauses are nested, and provisos can change the meaning of a sentence. Contract datasets such as CUAD [4] and examination datasets such as JEC-QA [5] show the need for domain-specific legal QA.

For Indian legal NLP, ILDC supports judgment prediction [6], and InLegalBERT provides a domain-adapted encoder [7]. IndicIPR-QA provides the extractive benchmark used here [8], and a companion manuscript studies domain-adaptive pretraining on this benchmark [9]. Domain-adaptive pretraining is known to help on in-domain tasks [10, 11]. These resources are text-centred: the model reads a passage and predicts a span. Recent legal benchmarks broaden this setting. LegalBench contains 162 expert-designed tasks covering six types of legal reasoning [12]. LLeQA studies long-form statutory QA with a retrieve-then-read pipeline and expert-annotated questions [13]. These benchmarks are complementary to the extractive setting studied here rather than direct replacements for it.

Groundedness is also becoming a separate evaluation target for legal QA. Recent work evaluates whether generated answers are supported by their supplied contexts, using similarity and natural-language-inference methods [14]. This distinction is relevant to the present study because a graph can change an answer without providing valid legal support for that change.

B. Legal ontologies and knowledge graphs

LKIF gives core legal concepts [15], LegalRuleML formalises legal rules [16], EuroVoc gives a multilingual vocabulary [17], and PROLEG models legal reasoning in logic programs [18]. OWL provides the representation framework [19]. These show the value of structure, but none covers Indian IPR statutes, and none is linked to a QA benchmark.

Most published legal KGs report size, such as the number of nodes, edges, and relation types, rather than conformance. A graph can be large and well connected while still violating its own schema. Recent legal KG construction has also used knowledge-enhanced LLMs, but the reported evaluation focuses on extraction accuracy and graph contents rather than a graph-level ontology audit [20]. This paper therefore develops an explicit conformance instrument.

C. Knowledge graphs for QA

Entity-aware encoders such as ERNIE [21] and K-ADAPTER [22] inject knowledge into language models. Subgraph retrieval methods select evidence for a question [23–26], and multi-hop systems such as PullNet retrieve linked evidence [27]. Recent work combines KG restructuring, LLM training, and subgraph retrieval for complex KGQA [28, 29]. Embedding methods including TransE [30], DistMult [31], and TuckER [32] score triples, and R-GCN extends message passing to typed relations [33]. These methods serve as components of the system rather than as new contributions.

Recent legal RAG benchmarks make retrieval quality an explicit evaluation target. LegalBench-RAG evaluates precise snippet retrieval in legal documents [34], while CLERC couples legal case retrieval with retrieval-augmented analysis generation [35]. A reasoning- focused benchmark further evaluates retrieval together with downstream legal QA and reports substantial difficulty for existing retrievers [36]. These studies motivate evaluating the graph, retrieval stage, and reader separately in this work.

Reported KG gains in this literature are often small and are measured against a single baseline. Sections X-E and XI-C show why both the strength and the composition of that baseline matter.

III. DATASET AND TASK

Every experiment uses the same benchmark and task definition. This section introduces IndicIPR-QA, reports the distribution of questions across the seven Acts, defines the extractive task and its metrics, and measures how much room the task leaves for a knowledge graph to contribute anything.

A. IndicIPR-QA

This work uses IndicIPR-QA [8], which contains 4,462 extractive QA records from Indian IPR statutes. The dataset has 3,710 training records, 408 validation records, and 344 test records. Each record contains a question, a statutory context, an answer span with offsets, and an Act label. The splits are made at the statute-section level, and no context or section in the test split occurs in training.

TABLE I
HEADROOM FOR EXTERNAL KNOWLEDGE IN INDICIPR-QA. THE ANSWER IS INSIDE THE PASSAGE THE READER IS GIVEN IN EVERY RECORD OF EVERY SPLIT, AND THE QUESTION USUALLY IDENTIFIES THE PROVISION ITSELF.

Split	Records	Answer in c	Section named	Words in c
Train	3,710	100.00%	93.21%	162.8
Val	408	100.00%	90.93%	270.2
Test	344	100.00%	94.48%	145.8

The answers are long. In the test split, the mean gold answer contains 39.9 words, which is about an order of magnitude longer than the answers in SQuAD. This affects decoding, as shown in Section X-D.

B. Act distribution

The test set covers all seven Acts, but not equally. It contains 88 Trade Marks records, 69 Patents records, 51 Geographical Indications records, 47 Semiconductor Layout-Design records, 40 Plant Varieties records, 25 Copyright records, and 24 Designs records. Aggregate scores are therefore influenced most by Trade Marks and Patents. Smaller subsets, such as Designs, have less statistical weight.

C. Task

Given a question q and context c , the task is to predict an answer span a from c . IPRon also retrieves a subgraph G_q for each question and may use it to rank candidate spans. The evaluation reports exact match (EM), token F1, and character F1. Character F1 is included because legal answers often differ only in punctuation or clause markers.

D. How much room the task leaves for external knowledge

A knowledge graph is useful to a QA system when a fact must be supplied from outside the passage, or when evidence must be joined across passages. Neither condition is guaranteed by a benchmark simply because the domain is legal, and a null result means very little if the benchmark never gave the graph anything to do. That premise is therefore measured before it is relied upon.

Table I gives the measurement. The gold answer occurs verbatim inside the supplied context in all 4,462 records, which follows from the extractive construction of the benchmark and means no external fact is ever required. The question names its own statutory section in 94.48% of test records, and names a section, subsection or clause in 95.93%, so the retrieval target is usually given rather than inferred. The passage itself averages 146 words.

The residual error is the second half of the argument, because headroom that exists is not necessarily headroom a graph can occupy. On the test split the reader, in the condition reported in Section XI-A without re-ranking, answers 170 of 344 exactly and a further 61 within 0.9 F1. Only 50 predictions, 14.5%, overlap the gold answer by less than half, and these are the only cases in which a missing fact is a

plausible explanation at all; the rest differ from the gold span by where it was cut or how it was worded.

What follows from this needs stating carefully, because the obvious inference is stronger than the evidence supports. Answerability does not require the graph: no external fact is ever missing, so the class of benefit that consists of supplying a fact, or of joining evidence across passages, has no room to appear. It does not follow that a graph cannot help at all. A graph may still disambiguate an entity, favour one candidate span over another, or supply an inductive bias to a reader that is weak for reasons unrelated to missing knowledge, and Section VI-C measures exactly that and finds +1.75 F1 against a weak reader.

The defensible statement is therefore narrower: this benchmark provides limited structural headroom for the benefits that motivate legal KGs in the first place, while leaving room for span-selection effects, so any QA result reported here is a result about extractive benchmarks of that shape. That is a weakness when the goal is to show a KG working for the reasons usually claimed, and a strength when the goal is to audit how KG benefit gets measured, because the mechanisms that remain available are narrow enough to be identified individually.

IV. IPRON: ONTOLOGY AND GRAPH

This section describes IPRON. It first explains why the graph needs a declared schema, then gives the classes and properties in that schema, and then describes the extraction pipeline that turns statutory text into edges. The last part describes connectivity repair, which improves the graph structure but also has a cost that the conformance measure reveals.

A. Design rationale

The extracted triples could have been stored without a schema. An ontology was chosen for two reasons. First, the same legal concepts recur across Acts, and a user should be able to query them as legal objects: to ask “which authority grants this right?” and get an answer based on entity types rather than on matching strings. Second, an ontology makes the graph checkable. Declared connection rules make it possible to measure whether the graph obeys them. This second reason matters most because the check became the most useful part of the study.

The provenance of the schema should be stated, because it bears on how the conformance figure is read. The classes and properties were authored for this work from concepts and relations that recur across the seven statutes, rather than adapted from a published legal ontology such as LKIF [15] or LegalRuleML [16], which are discussed in Section II as related work rather than used as a base. The schema was not independently validated by a legal expert. Conformance is therefore agreement with a schema the same authors declared, which is what makes it a check on internal consistency rather than on legal correctness — a limitation revisited in Sections V-D and XIV.

B. Schema

IPRON declares 13 classes: Act, Section, IPRightType, IPHolder, Authority, Procedure, Right, Obligation, Penalty, Remedy, Definition, Infringement, and Entity. Entity is the superclass of the others and is also the fallback type for a phrase that cannot be classified more precisely. It also declares 23 object properties, which this paper separates into two groups throughout. Semantic properties state legal facts. Examples include grantedBy (domain IPRightType, range Authority), definesAs (domain Section, range Definition), hasPenalty, hasRemedyFor, imposesObligation, and revokedBy. Structural properties record where something appears rather than what is legally true: mentionedIn, co_occurs_with, relatedTo, relatedToCore, and sameAs. These help retrieval, but they do not assert legal facts.

Keeping the groups apart prevents errors from being hidden. A graph in which 91% of the edges are structural can look highly conformant even when almost none of its legal claims satisfies the schema. Any ontology-guided KG should report the two layers separately.

C. How the graph is built

The graph is built in three stages. First, recurring legal terms are taken from the training corpus with per-Act KeyBERT scoring [37] and mapped to an ontology class. This gives the graph an anchor vocabulary, so later stages can attach new facts to known concepts rather than to arbitrary strings. Next, hand-written patterns are applied to common and reliable statutory forms: definitions (“...means ...”), terms of protection, grants of rights, revocation, and penalties. These rules aim for high precision and accept low coverage. They contribute 169 distinct triples.

To cover the remaining text, each sentence is parsed with a dependency parser to identify subject, verb, and object candidates. An ontology relation is then assigned to every candidate using cosine similarity between an InLegalBERT encoding of the candidate and precomputed relation descriptions, with a verb-pattern method as a fallback. This stage contributes 1,091 distinct triples. It makes 5,133 similarity assignments against 614 verb-pattern fallbacks, at a mean cosine similarity of 0.699.

The two extractors agree on exactly one triple. They therefore give each other almost no corroboration. This warning sign is discussed in Section V.

These counts are of distinct subject–relation–object triples. The construction-strategy comparison in Section VI counts the same three graphs by edge–relation record, which is the unit the conformance measure uses, and so reports 210, 1,167 and 1,332 for the same runs. Both countings are given because the extraction discussion is about what was found and the conformance discussion is about what is checked.

D. Connectivity repair and its cost

The extracted graph has 2,083 nodes and 5,574 edges, split across 508 components, with 340 isolated nodes. Retrieval over a graph in this shape often fails, because a question can land on a node that has no useful neighbourhood.

TABLE II
LEGAL GRAPH BEFORE AND AFTER CONNECTIVITY REPAIR.

Metric	Before	After
Nodes	2,083	2,383
Edges	5,574	17,560
Connected components	508	23
Isolated nodes	340	0
Largest component	1,359	2,336

Connectivity is repaired with five mechanisms: section anchors (201 nodes and 11,051 edges), answer mining (47 nodes and 51 edges), definition repair (52 nodes and 57 edges), cross-Act normalisation (4 edges), and conservative bridges (823 edges). After repair the graph has 2,383 nodes, 17,560 edges, 23 components, and no isolated nodes (Table II).

The repair makes the graph far easier to search, but it is not the same thing as better extraction, and the two should not be confused. Section anchoring alone performs 11,051 edge insertions, all of them mentionedIn, which only records that a phrase appears in a section; 10,909 survive as distinct edges, the rest landing on pairs that were already joined. Edges like these help locate evidence. They state no legal facts. This is why the paper reports the two layers separately.

V. MEASURING CONFORMANCE

A graph can be large, well connected, and still disagree with the schema published alongside it. This section defines a test for that disagreement. It states the three checks, reports their results on IPRON, asks what causes the result, and then asks how much of the cause is the ontology’s own doing.

A. Definition and procedure

Publishing an ontology with a graph implies that the graph instantiates the ontology. The original tooling did not check this property, so a validator was developed. The validator asks three questions about every edge. Each question is stricter than the previous one.

- 1) C1: Is the relation declared? Does the edge use a property that exists in the ontology?
- 2) C2: Are both endpoints typed? Do the subject and object carry a declared class?
- 3) C3: Are the domain and range satisfied? Does the subject’s class match the property’s declared domain, and the object’s class its range?

An edge is conformant only if it passes all three checks. The validator uses subsumption rather than string equality. Because Procedure is a subclass of Entity, a Procedure node satisfies a property that declares domain Entity. Testing only the exact class name would understate conformance for properties declared over a superclass.

Algorithm 1 states the procedure. Two details are easy to get wrong and both changed the reported numbers. First, a name can be declared both as a datatype property and as an object property. The term of protection is a literal in principle, but the pipeline emits hasTermOf between two nodes, so the ontology declares both forms. Testing the datatype table first

Algorithm 1 Ontology conformance validation

Require: Graph $G = (V, E)$; ontology O with classes C , object properties P , datatype properties D

Ensure: Per-layer counts of C1, C2 and C3 outcomes

```

1:  $A \leftarrow \text{ANCESTORS}(C)$  ▷ closure of subClassOf
2: for all edges  $(u, v)$  with relation  $r$  in  $E$  do
3:   if  $r \in D$  and  $r \notin P$  then
4:     record MISUSEDDATATYPE; continue
5:   end if
6:   if  $r \notin P$  then ▷ C1 fails
7:     record UNDECLARED; continue
8:   end if
9:    $\ell \leftarrow \text{STRUCTURAL}$  if  $P[r]$  is structural else SEMANTIC
10:   $c_u \leftarrow \text{CLASS}(u)$ ;  $c_v \leftarrow \text{CLASS}(v)$ 
11:  if  $c_u \notin C$  or  $c_v \notin C$  then ▷ C2 fails
12:    record UNTYPEDENDPOINT in layer  $\ell$ ; continue
13:  end if
14:  if  $\text{SAT}(c_u, P[r].\text{dom}, A)$  and  $\text{SAT}(c_v, P[r].\text{rng}, A)$  then
15:    record CONFORMANT in layer  $\ell$  ▷ C3 holds
16:  else
17:    record domain or range violation in layer  $\ell$ 
18:  end if
19: end for
20: return counts, reported separately for each layer

21: function  $\text{SAT}(c, d, A)$  ▷ subsumption, not equality
22:   if  $d$  is undeclared then
23:     return true
24:   end if
25:   if  $c$  is undeclared then
26:     return false
27:   end if
28:   return  $d \in A[c]$ 
29: end function

```

sent all 190 hasTermOf records to a “misused datatype” bucket and excluded them from the semantic layer altogether, which reported the layer as 1,392 records at 19.54% rather than 1,582 at 19.28%. Object properties must therefore be tested first; 33 of those 190 records are conformant. Second, the counters are kept per layer, so a graph cannot hide a broken semantic layer behind conformant structural edges.

B. Result

Table III reports the results. Overall conformance is 92.76%. Structural edges dominate this value because they are 100% conformant by construction. The semantic layer is the more important measure. Its conformance is 19.28%: only 305 of the 1,582 edges that assert legal facts satisfy their declared types.

The graph contains 17,560 edges, and the audit reports 17,638 records. The two numbers differ because the unit of the audit is the edge–relation pair rather than the edge: 65 edges carry more than one relation label, 52 of them two and 13 of them three. Each label is checked on its own, because each one is a separate claim about the same pair of nodes.

TABLE III

CONFORMANCE BY LAYER, OVER THE 17,638 EDGE-RELATION RECORDS IN THE 17,560-EDGE GRAPH. THE SEMANTIC LAYER IS THE MEANINGFUL FIGURE: THESE ARE THE RECORDS THAT ASSERT LEGAL FACTS.

Layer	Records	Conformant	Rate
Structural	16,056	16,056	100.00%
Semantic	1,582	305	19.28%
All records	17,638	16,361	92.76%

Figure 2 shows the result for each property. Every property defined by hand is fully conformant: `hasSection`, `governedBy`, `regulatedBy`, `regulates` and `sameAs` are all at 100%. Every property populated by automatic extraction has mixed quality. `imposesObligation`, `definesAs`, `grantsRight` and `hasRemedyFor` are at 0.0%. The two largest properties, `revokedBy` and `grantedBy`, are at 0.8% and 1.4%, respectively, and `hasTermOf`, the third largest, is at 17.4%. Smaller properties such as `submittedTo`, `governs`, `hasPenalty` and `infringedBy` have higher rates, from 40% to 75%, but contain fewer edges.

C. Diagnosing the residual

A low number is only useful when its cause is known. Three explanations were tested, each of which would have pointed to an easy fix. All three are computed by a released diagnostic script that reuses the validator’s own primitives, so the analysis cannot drift from the measure.

Was the ontology incomplete? At the start, 90.76% of edges used relations that had not been declared, among them `mentionedIn`, `co_occurs_with`, and the other structural links. Those relations are real and useful, so they were declared, `Entity` was added as a class, and the subclass hierarchy was specified. Overall conformance rose from 1.53% to 92.76%. The semantic layer did not change. This was a gap in the documentation, not a problem in the data.

Were the endpoints simply untyped? 1,362 nodes carried the `Entity` fallback type, and a lexicon-based typer gave specific classes to 490 of them. Reverting those 490 assignments moves semantic conformance from 18.46% to 19.28%, a gain of 0.82 percentage points from 13 repaired records. Every other record keeps the outcome it already had. The reason so little changes is that `Entity` is itself a declared class, so a fallback-typed endpoint never fails the C2 test: it passes C2 and then fails C3 on the declared domain or range. The fallback was hiding the type errors rather than causing them, and giving a node a better type exposes the error instead of removing it.

Was the argument order simply reversed? The ontology declares `grantedBy` as `right` $\rightarrow$ `authority`, and extraction often produces `authority` $\rightarrow$ `right` instead. If reversed arguments were the main problem, flipping the non-conformant triples would fix them. Flipping recovers 20 of the 1,277 non-conformant records, or 1.26 percentage points. The reason it recovers so few is that 1,257 records, 79.46% of the semantic layer, violate their declared types in both directions.

That leaves relation assignment itself. The extraction log shows the mechanism, on examples whose intended relation can be read off the source phrase directly.

TABLE IV

CONFORMANCE OF THE SEMANTIC LAYER UNDER RELAXED DECLARATIONS ($n = 1,582$ RECORDS). THE ORACLE IS AN UPPER BOUND, NOT A PROPOSAL.

Declaration	Conformant	Rate
As declared (reported)	305	19.28%
Domain enforced, range relaxed	413	26.11%
Range enforced, domain relaxed	542	34.26%
Oracle: modal class per position	1,265	79.96%

Triple	Assigned	Cosine
central government $\rightarrow$ may appoint	<code>hasTermOf</code>	0.76
any building $\rightarrow$ having	<code>appliesTo</code>	0.75
application $\rightarrow$ is made to	<code>grantedBy</code>	0.82

The third row shows the mechanism. The phrase “is made to” points to `submittedTo`, but the classifier picked `grantedBy` at a similarity of 0.82. The root cause is that the classifier has no reject option: every candidate above the 0.60 threshold is given some relation. Since the mean similarity across the corpus is 0.699, almost nothing falls below the threshold, and the top-scoring relation can be close to arbitrary. Mean-pooled encoder embeddings make this worse, because they are anisotropic and squeeze the cosine values into a narrow high band, which flattens the ranking further.

D. How much of the residual is the ontology’s own fault

Every number above compares extracted arguments against types this paper declared. A low score therefore admits a second reading, and it is the reading a reviewer should press: perhaps the extractor is adequate and the ontology declared its argument types too narrowly. That reading is not idle here. `imposesObligation` declares domain `Section`, and the extractor produces a phrase typed `Authority` in subject position 146 times. “The registrar imposes an obligation” is a sensible legal statement that the declared domain forbids.

Table IV bounds the possibility. Each row relaxes part of the declaration and rescores the same 1,582 records. The last row is an oracle: for every property, domain and range are refitted to the most common class actually observed in that argument position. The oracle is fitted to the graph it is scoring, so it is not a proposed ontology; it is the ceiling that any assignment of domains and ranges could reach on this data.

Taken alone, the oracle’s 79.96% would say the declarations are most of the problem. It does not survive inspection. The oracle reaches that figure mainly by selecting `Entity` as the modal class, and `Entity` is the superclass of everything, so the constraint becomes vacuous: `revokedBy`, `definesAs` and `appliesTo` all reach 100% that way. The bound is real but it is bought by removing the test.

The failure composition is the more informative statistic. Of the 1,277 non-conformant records, 831 (65.1%) involve an endpoint carrying the generic `Entity` fallback, and only 446 (34.9%) are mismatches between two specific declared classes. Two thirds of the residual is therefore a typing failure in the pipeline rather than a modelling choice in the ontology, which is consistent with Section V: the lexicon-based typer left 872

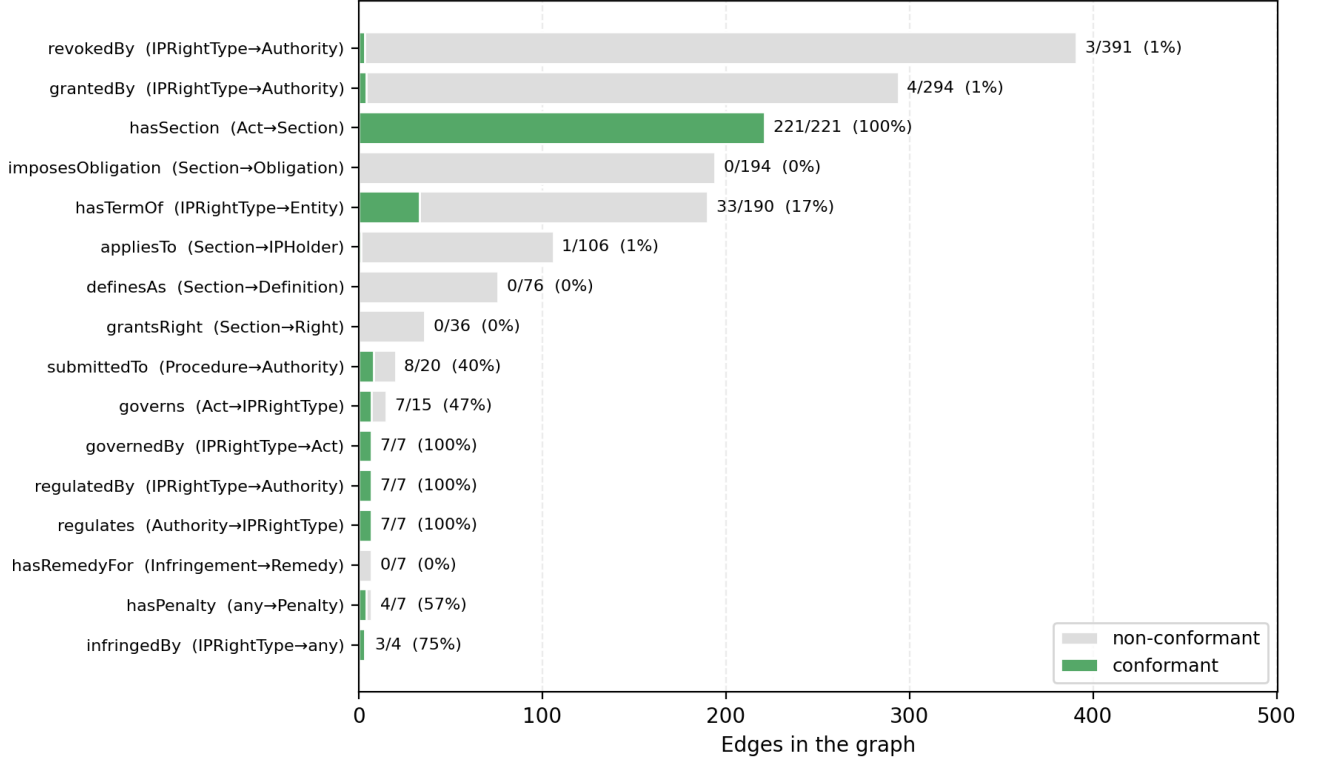


Fig. 2. Each semantic property with its declared domain and range, showing how much of what the graph contains actually satisfies that declaration.

nodes generic, and an edge with a generic endpoint cannot satisfy a property declared over a specific class.

The honest conclusion is mixed, and the paper reports it as such. Two properties have a defensible alternative declaration, *imposesObligation* with domain *Authority* and *grantsRight* with domain *IPRightType*, and their conformance under the oracle rises to 75.3% and 36.1% respectively; 10 of the 16 populated semantic properties are declared differently from their modal observed class. So 19.28% understates how compatible this graph is with alternative declarations of its own schema. It does not measure legal correctness in either direction: the oracle is fitted to the graph it scores, and most of its increase comes from relaxing the constraint rather than from the data being right. Nor does it follow that the graph is sound, because the dominant defect is that two thirds of the conformance failures involve an endpoint the pipeline never assigned a specific class, leaving it with the Entity fallback.

VI. CONSTRUCTION STRATEGY, CONFORMANCE, AND DOWNSTREAM QA

The result in Section V concerns one extractor. It does not by itself describe ontology-guided construction in general. Therefore, the corpus, ontology, and conformance measure were held fixed, and only the construction strategy was changed. The section then asks the question that matters downstream: whether a better graph produces better answers.

This study samples 300 statutory sentences and compares four strategies. The sample is drawn uniformly at random, without replacement and with seed 42, from the 2,131 distinct

TABLE V
CONSTRUCTION STRATEGY VERSUS CONFORMANCE, ON THE SAME CORPUS, ONTOLOGY AND MEASURE. S4 SAMPLES 300 SENTENCES; S5 REPEATS THE SAME PROMPT OVER ALL 2,131. TRIPLES PER SENTENCE NORMALISES FOR THE DIFFERENT SENTENCE COUNTS. THESE RATES ARE MEASURED AT THE EXTRACTION STAGE AND ARE NOT COMPARABLE TO THE 19.28% REPORTED FOR THE COMPLETED GRAPH; SEE THE TEXT.

Strategy	Triples	/sent.	Conf.	Rate
S1 rule-based	210	0.099	46	21.90%
S2 OpenIE + cosine	1,167	0.548	69	5.91%
S3 hybrid (S1+S2)	1,332	0.625	71	5.33%
S4 LLM (Qwen2.5-7B)	298	0.993	127	42.62%
S4 LLM (Gemma3-12B)	285	0.950	127	44.56%
S5 LLM full corpus	1,870	0.878	675	36.10%

sentences obtained by splitting the deduplicated statutory contexts and keeping those between 60 and 400 characters; the length filter removes headings and fragments that carry no extractable relation. All strategies see the same 300 sentences, so the comparison holds the input fixed. The fourth strategy is a strong baseline. It uses a local LLM that receives every relation, its declared domain and range, and every class. The prompt also states explicitly that the model may return an empty list when no relation applies.

One point of comparability needs stating before the results, because two conformance figures in this paper are easy to conflate. The 5.33% recorded here is measured over the 1,332 triples the hybrid pipeline extracts from the 300 sampled sentences, at the extraction stage. The 19.28% reported elsewhere is measured over the 1,582 semantic records of the

completed graph, after the section anchoring, answer mining and bridging of Section IV have added edges the extractor never produced. The two rates describe different populations at different pipeline stages; only the rows within Table V are directly comparable with each other.

Table V shows four main results.

The simplest extractor is the most conformant of the three pipeline variants. Rule patterns reach 21.90% conformance. Adding OpenIE lowers this value to 5.91%. Combining the two methods gives 5.33%, which is lower than either method alone. The additional coverage from OpenIE comes with a fourfold loss in conformance.

Both LLMs outperform the pipeline on both measures. They produce about 1.6 times more facts per sentence and achieve about eight times higher conformance. In this comparison, prompted extraction does not trade coverage for precision. It performs better on both measures.

The difference is visible in the rejection behaviour. Gemma3-12B returned an empty list for 91 of 300 sentences, and Qwen2.5-7B returned an empty list for 51. The models can decline when no relation applies. The pipeline cannot do this, and this missing capability explains much of the gap.

The advantage is not an artifact of sample size. S4 extracts a few hundred triples while the pipeline graph holds 1,332, so a reviewer may reasonably suspect that a small, carefully chosen set is simply easier to keep type-correct. S5 removes that concern by running the identical prompt over every sentence in the corpus. The result is 1,870 triples, a larger set than the pipeline produces, at 36.10% conformance, still 6.8 times the pipeline’s rate. Scale does cost something: conformance falls from 44.56% on the sample to 36.10% on the full corpus. This paper reports that drop rather than the sample figure alone because the full corpus number is the one a practitioner would actually obtain. It leaves the comparison unchanged in direction and in order of magnitude. The rejection behaviour also holds up at scale, with 787 of 2,131 sentences (36.9%) declined, close to the 30.3% observed on the sample.

These results do not establish legal correctness. At 44.56%, more than half of the LLM output is still non-conformant. In addition, conformance measures type correctness rather than legal truth. A triple can satisfy its domain and range and still misread the statute. Human annotation is required to establish legal correctness, and that annotation is in progress.

A. A provisional check on what conformance predicts

Pending that annotation, a sample of 299 semantic triples was labelled independently by two local LLMs as legally correct or incorrect against the source sentence. This is a proxy for the human study, not a substitute for it, and it is reported here only because it bears directly on how far conformance can be trusted.

The two annotators agree on the direction of the effect and disagree on its level. For both, type-conformant triples are judged legally correct significantly more often than non-conformant ones: 64.56% against 43.12% ($\chi^2 = 9.83$, $p = 0.002$) for one annotator, and 97.50% against 82.19% ($\chi^2 = 10.35$, $p = 0.001$) for the other. But their overall correctness

TABLE VI
THE LLM GRAPH, ACTUALLY USED. ALL LLM CONDITIONS USE THE STRING-HEURISTIC KG SCORER, BECAUSE THE LEARNED EMBEDDINGS WERE TRAINED ON THE PIPELINE GRAPH’S ENTITY VOCABULARY AND CANNOT SCORE A DIFFERENT GRAPH. ALL DOWNSTREAM QA VALUES USE THE VALIDATION-SELECTED DECODER CAP OF 200.

Graph	Edges	Conf.	Changed	Δ F1
Pipeline (full)	17,560	19.28%	4	+0.00
LLM, Qwen2.5-7B	193	37.88%	1	+0.17
LLM, Gemma3-12B	246	35.34%	1	+0.17
LLM, Gemma3-12B full	1,440	30.94%	1	+0.17

rates are far apart, at 145 and 258 of 299, and their agreement is only fair (Cohen’s $\kappa = 0.22$, 60.2% raw). No level should be read off these numbers.

Two things follow, and only two. Both model-based annotators show an association between conformance and their own legal-correctness judgments, which suggests conformance may carry information about correctness rather than being an unrelated formal property; human legal annotation is required to establish whether that relationship is real and to estimate its size. And conformance is not necessary for legal correctness: on both annotations a large group of triples is judged legally correct while failing its declared types, 94 and 180 respectively. Neither is it sufficient, since 35.44% and 2.50% of conformant triples are judged incorrect. Conformance is therefore neither necessary nor sufficient for legal correctness, and should not be read as a per-triple correctness verdict in either direction.

B. Downstream effect of a more conformant graph

A more conformant graph is useful only if it improves downstream QA. Therefore, a graph was built from the LLM triples, and the same entity-linking and re-ranking procedures were applied.

Building a graph is stricter than counting triples. Self-loops, undeclared relations, and duplicates are removed. Of the 285 raw Gemma3-12B triples, 6 use an undeclared relation and 3 are self-loops. After deduplication, 249 semantic records remain over 246 distinct edges. Graph-level conformance is therefore 35.34%, compared with 44.56% on the raw triple list. The same gap appears at corpus scale, where 1,870 raw triples reduce to 1,503 records over 1,440 edges, and 36.10% becomes 30.94%. Both values are reported to keep the two measures distinct. The Conf. column of Table VI is the graph-level figure in every row, so the rows are comparable with each other even though they are stricter than Table V.

Table VI shows the result. The sampled Gemma3 graph is 71 times smaller than the pipeline graph and is roughly twice as conformant. It also produces a small positive change, whereas the full pipeline produces no change. The effect is not significant: the gain is +0.169 F1, and a bootstrap interval over records spans zero at $[+0.000, +0.507]$ because a single prediction out of 344 changes. Growing the LLM graph almost six-fold, to 1,440 edges at 30.94% conformance, also changes one prediction and moves F1 by the same +0.169.

The natural reading is that better extraction produces a better graph but not better answers. That reading should be resisted,

and this subsection is reported as an illustration rather than as evidence. The comparison was run against a reader at the top of this benchmark, on a task where the answer is already in the passage in every record (Section III-D). A reader with no measurable room to improve cannot distinguish a graph that would have helped from one that would not; the same comparison against a weak reader is made in Section VI-C. Two further confounds keep this from being a clean test of conformance. All LLM conditions use the string-heuristic KG scorer, so the scorer is not held fixed across the rows in the way the extractor is. And the graphs differ in size, though less than the edge counts suggest: retrieval discards structural relations, so what it sees is the semantic layer, where the pipeline offers 1,582 records against the full-corpus LLM graph’s 1,503, a difference of 5%. The controlled version of this comparison is Section VI-C.

C. RQ4: does ontology conformance change downstream QA?

The comparison above confounds conformance with size, with the scorer, and with a reader that has no room to improve. This section removes all three. The pipeline’s semantic layer (1,504 edges, 19.28% conformant) is compared against the full-corpus prompted graph (1,440 edges, 30.94%), with the pipeline layer randomly subsampled to exactly 1,440 edges over five seeds so that size is matched by construction rather than by argument. A third graph is added: the same pipeline output with the declared types enforced at construction time, so that a candidate triple is stored only if its subject satisfies the property’s domain and its object satisfies its range. That graph is 100% conformant by definition and retains 304 of the 1,504 edges, close to the 300 to 500 triples this construction was expected to yield.

Everything downstream is shared: the same entity linker, the same string-heuristic scorer, the same α and β , the same candidate spans from the same reader, and the same 344 records. Learned embeddings are excluded deliberately, since they can only score one of the graphs. Each condition is run against a weak reader, which has room to improve, and the strong reader. Intervals are paired percentile bootstrap over per-record F1, 10,000 resamples.

Table VII answers RQ4 in three steps.

First, the design has power. Against the weak reader the graph is worth +1.75 F1 with an interval that excludes zero, so re-ranking on a retrieved subgraph does help a reader that has room to improve, and this experiment can detect it.

Second, no difference associated with conformance is detectable. Holding size and scorer fixed, replacing the low-conformance graph with one 1.6 times more conformant is worth -0.07 F1, with an interval spanning zero. Almost all of the benefit is from having a graph at all; almost none of it is from the graph satisfying its schema. The variable the contrast is built around is conformance, not legal correctness: the only correctness labels available are the model annotations of Section VI-A, whose inter-annotator agreement is $\kappa = 0.22$, so no claim is made that the more conformant graph is the more legally accurate one. The five subsamples vary by 0.04

TABLE VII
ONTOLOGY CONFORMANCE AT MATCHED SIZE AND SCORER ($n = 344$).
LOW-CONFORMANCE ROWS ARE MEANS OVER FIVE SIZE-MATCHED
SUBSAMPLES. THE GRAPH EFFECT ON THE WEAK READER IS THE
POSITIVE CONTROL: IT ESTABLISHES THAT THIS DESIGN CAN DETECT A
GRAPH MAKING A DIFFERENCE.

Condition	Weak reader	Strong reader
No graph	28.83	84.88
Low conformance, size-matched	30.65 ± 0.04	84.77 ± 0.00
High conformance, same size	30.57	85.05
Graph effect (high – none)	+1.75	+0.17
95% CI	[+0.79, +2.87]	[+0.00, +0.51]
Conformance contrast (high – low)	-0.07	+0.28
95% CI	[-1.29 , +1.08]	[-0.00 , +0.71]
Type-enforced, 100% conformant	29.09	84.88
Low conformance, matched to it	30.08 ± 0.54	84.81 ± 0.05
Enforcement effect	-0.94	+0.12
95% CI	[-1.80 , -0.19]	[-0.03 , +0.31]

F1, so this is not an artifact of which 1,440 edges were drawn, and both graphs change nearly the same number of predictions, 19 against 20, so the more conformant graph is not intervening more often.

Third, and least expected, obtaining conformance by hard type filtering is actively harmful. The type-enforced graph is perfectly conformant and scores -0.94 F1 against an equally sized random draw from the unenforced layer, with an interval of [-1.80 , -0.19] that excludes zero. Enforcement keeps only a fifth of the edges, and reach falls from 344 of 344 questions to 326. The coverage it discards is worth more to downstream QA than the increase in schema conformance it buys. On the strong reader the enforced graph changes no predictions at all.

This strengthens the near-zero correlation of Section XI-D1 from a passive observation into a controlled contrast, though not into a clean intervention. There, questions whose retrieved evidence was more conformant were not answered better ($r = 0.021$). Here the graph is chosen rather than observed, twice, with size and scorer held fixed, and the more conformant graph is never better: once within noise, once significantly worse. The contrast is not a single-variable manipulation — the graphs also differ in content and topology, and the enforced graph differs in coverage — so what the two rows jointly support is that no route from higher conformance to better answers is visible in this architecture, not that conformance has been isolated and shown inert. Conformance is a property of the graph as a document; on this benchmark it is not a property this retrieval-and-re-ranking architecture rewards.

Two limits apply, and neither is cosmetic. The first is power: the interval on the conformance contrast, [-1.29 , +1.08], is wide enough to contain effects up to about 1.1 F1, so this rules out an effect the size of the graph effect but not a small one. The second is identification. Size matching makes the two graphs comparable in scale, but they are still differently constructed, and they differ in which triples exist, in entity and relation distribution, in topology and in lexical overlap with the questions. Conformance is the property being contrasted, not the only property that varies. The enforcement row is cleaner in that respect, since it filters the pipeline’s own output and so holds the extractor fixed, but it manipulates two

TABLE VIII

THE SAME VALIDATOR APPLIED TO INDEPENDENTLY PUBLISHED LEGAL KGs. FAILURE COUNTS SHOW C1, C2 AND C3 RESPECTIVELY. THE TWO GRAPHS ARE NOT COMPARABLE IN KIND; SEE THE CAVEATS IN THE TEXT.

Graph	Edges	C1/C2/C3 fail	Conf.
EVA-KG, bottom-up	1,976	146 / 1,830 / 0	0.00%
PreJust4Womans, LLM	74	11 / 0 / 4	79.73%
IPROn pipeline	1,332	—	5.33%
IPROn prompted	1,870	—	36.10%

things at once: perfect conformance is obtained by discarding four fifths of the edges. What that row establishes is that hard type filtering under substantial coverage loss hurts, not that conformance itself does. A type-enforced extractor that recovered the discarded content by other means might behave differently, and building one is the natural next experiment.

The whole experiment was also run at the original cap of 150 (Section X-D), and the two effects that carry weight are unchanged to two decimals: the graph is worth +1.75 F1 with interval [+0.79, +2.87], and type enforcement costs −0.95 F1 with interval [−1.82, −0.21]. The conformance contrast moves from −0.07 F1 here to +0.19 at the smaller cap, changing sign while both intervals continue to span zero. That instability is itself the clearest statement of the result: an effect whose sign depends on the decoder cap is an effect this benchmark cannot resolve, which is what “no detectable difference” means here.

D. Applying the validator to independently published legal KGs

The preceding analysis concerns graphs built for this study. A reader may reasonably ask whether the validator is usable by others and whether low conformance is peculiar to this pipeline. The validator was therefore applied, unchanged, to a legal knowledge graph published by a different group.

The resource of d’Amato et al. [38] is used as a useful test case because the same authors publish two graphs over the same corpus of European Court of Human Rights decisions on violence against women: one built bottom-up, and one built by prompting an LLM. That is the comparison of Table V, carried out independently, in a different jurisdiction and a different area of law. A generic version of the validator reads their graphs. It takes `rdfs:domain`, `rdfs:range`, `rdfs:subClassOf` and `rdf:type` directly from any OWL or RDFS graph and applies the same three checks with the same subsumption rule.

Table VIII reports the results. Three narrow observations follow.

First, the validator runs on a graph it was not designed for, and it finds real problems. Their published ontology does not parse at all: it uses `xsd:string` without declaring the `xsd` prefix, so no compliant parser will load it, and the missing line had to be added before its content could be tested. One instance file also contains an unterminated string literal and is dropped by the parser. Neither defect is subtle, and neither would survive a conformance run. This case demonstrates that basic schema and syntax defects can survive peer review and

publication, and illustrates the practical value of an automated check. It is one resource by one group, and no claim is made here about the legal-KG literature in general.

Second, the ordering in Table V appears again in an independent dataset. Their LLM-built graph is far more conformant than their bottom-up graph, just as prompted extraction is far more conformant than the pipeline in this study. That experiment was designed independently.

Third, the failure modes differ, which is why the C1/C2/C3 decomposition is reported instead of a single number. The residual in IPROn is dominated by C3: the arguments do not satisfy their declared types. The other graph is dominated by C2: 1,830 edges connect endpoints that carry no declared type. This is not explained by links to Wikidata, where types could live remotely: only 8.52% of those C2 failures involve an external IRI.

Two caveats matter. The graphs are not comparable in kind: EVA-KG is largely document metadata, with 6,752 of its assertions taking literal values and only four properties declaring a class-valued range, whereas IPROn asserts legal facts. And their LLM graph is small, at 74 object-property edges, so 79.73% is an indicative figure rather than a reliable estimate. Thus, Table VIII is evidence that the instrument transfers, and that in this published resource the schema was not checked before release. It is neither a ranking of graph quality nor a claim about how often such checks are performed in the literature, which this paper has no basis to estimate.

VII. QA-LINK CONSTRUCTION

A graph built from statutes is not yet connected to a benchmark. Something has to tie each question and each answer to the nodes they talk about, or the retrieval step has no place to start. This section explains why these links are added, their types, and how much of the benchmark they reach.

A. Why link QA data into the graph

Legal KGs and QA datasets are usually built separately. This work joins them so that questions and answers become part of the graph rather than remaining external training rows. This allows the legal entities mentioned by each question, the section that supports each answer, and a graph-embedding training signal that includes QA structure in addition to extracted triples.

B. Link types and coverage

For each record, a question node and an answer node are created. They are connected with five relations: `asks_about` (question to legal entities), `has_answer` (question to answer), `from_section` (record to statutory section), `supported_by` (answer to supporting context), and `mentions` (question or answer text to entities).

The QA-linked graph has 11,307 nodes and 65,670 edges, of which 48,110 are QA links: 19,704 mentions, 15,020 `asks_about`, and 4,462 each of `has_answer`, `from_section` and `supported_by` (Table IX).

Three of these five relations occur once per record by construction. Their “100% coverage” is therefore definitional

TABLE IX
QA-LINKED GRAPH.

Item	Count
Legal nodes / edges	2,383 / 17,560
Augmented nodes / edges	11,307 / 65,670
Question nodes	4,462
Answer nodes	4,462
QA-link triples	48,110
mentions	19,704
asks_about	15,020
has_answer	4,462
from_section	4,462
supported_by	4,462

rather than an empirical result. Only mentions and asks_about are inferred from the data, and both are produced by string matching.

VIII. RETRIEVAL AND GRAPH EVIDENCE

Given a question, the system must decide which part of the graph is relevant and hand that part to the reader as evidence. This section describes how a question is matched to entities, how a subgraph is grown around them, and what counts as a multi-hop chain. The definition of a chain carries more weight than it first appears, because a permissive one inflates multi-hop coverage without adding any evidence.

A. Entity linking and subgraph retrieval

For each record, the question and context text are linked to graph entities using lexical normalisation, seed-vocabulary matching, section recognition, and embedding similarity. A beam search then starts from those entities to collect a subgraph and path chains, as set out in Algorithm 2.

Two design choices in that algorithm are easy to get wrong and change the reported coverage. First, the search walks in-edges as well as out-edges, so the frontier must advance to the neighbour in both cases. Advancing to the neighbour only on out-edges leaves the frontier standing still on every in-edge, and the chain is then emitted but cannot be walked. Second, each hop is recorded by its direction of travel (from and to) rather than by subject and object. A chain is connected exactly when $\pi_i.to = \pi_{i+1}.from$. Storing subject and object instead makes consecutive hops share an object, which made 89.4% of the recorded chains appear disconnected when they were not.

Lines 18, 20 and 22 are the backtracking guards. Because the search walks edges in both directions, without them it will step $A \rightarrow B$ and then $B \rightarrow A$ along the same edge and call the result a two-hop chain. Such a chain is one edge counted twice, not multi-hop evidence.

Across all splits, coverage is 99.75%. In the test split, every record links to at least one entity. Each record has, on average, 3.64 linked entities, 8.68 subgraph nodes, 9.35 subgraph edges, and 5.15 path chains (Table X).

Algorithm 2 Subgraph retrieval with backtracking guards

Require: Question q ; graph G ; topic entities T ; beam width b ; depth H

Ensure: Set of connected multi-hop chains $\mathcal{C}$

```

1:  $B \leftarrow \{m \mapsto \{[]\} : m \in T \cap V(G)\}$ ;  $\mathcal{C} \leftarrow \emptyset$ 
2: for  $h = 1$  to  $H$  do
3:    $K \leftarrow \emptyset$ 
4:   for all  $(n, \Pi) \in B$  do
5:     for all edges at  $n$ , in either direction, to  $m$  with
       relation  $r$  do
6:       if  $r$  is structural then continue  $\triangleright$  no legal
       content
7:       end if
8:        $K \leftarrow K \cup \{(n, r, m, \Pi)\}$ 
9:     end for
10:   end for
11:   score each  $k \in K$  by  $\cos(\text{ENC}(q), \text{ENC}(r))$ 
12:   keep the highest-scoring instance of each relation type
13:    $K \leftarrow$  top- $b$  elements of  $K$ 
14:    $B' \leftarrow \emptyset$ 
15:   for all  $(n, r, m, \Pi) \in K$  do
16:      $t \leftarrow \langle \text{from}=n, \text{to}=m, r, \text{inverse} \rangle$ 
17:     for all  $\pi \in \Pi$  with  $\pi \neq []$ , last hop  $p$  do
18:       if  $p.from = m$  then continue  $\triangleright$  immediate
       backtrack
19:       end if
20:       if  $\{p.from, p.to\} = \{n, m\}$  then continue  $\triangleright$ 
       edge reused
21:       end if
22:       if  $m$  already on  $\pi$  then continue  $\triangleright$  cycle
23:       end if
24:        $\mathcal{C} \leftarrow \mathcal{C} \cup \{\pi \cdot t\}$ ; append  $\pi \cdot t$  to  $B'[m]$ 
25:     end for
26:   end for
27:    $B \leftarrow B'$ 
28:   if  $B = \emptyset$  then break
29: end if
30: end for
31: return  $\mathcal{C}$ , with nodes ranked by similarity to  $q$ 

```

TABLE X
RETRIEVAL STATISTICS BY SPLIT.

Split	Coverage	Entities	Nodes	Chains
Train	99.70	3.35	7.92	4.72
Val	100.00	3.17	8.15	5.10
Test	100.00	3.64	8.68	5.15

B. What counts as a multi-hop chain

A path chain provides evidence of multi-hop structure only if it is a genuine path. The beam can traverse edges in both directions. Without safeguards, it can move from A to B and then return from B to A along the same edge, which creates a false two-hop chain. Without the guards, 75.2% of the “multi-hop” chains reuse one edge and 75.9% return to the starting node.

An extension is rejected if it returns to the node just left,

TABLE XI
LINK-PREDICTION METRICS, EVALUATED ON 2,000 VALIDATION TRIPLES.

Model	MRR	Hits@1	Hits@10
DistMult	0.4375	0.4090	0.4715
TransE	0.4298	0.3880	0.5095
TuckER	0.4087	0.3790	0.4520
InLegalBERT cosine (zero-shot)	0.3854	0.3810	0.3925

reuses an edge in either direction, or revisits a node already on the path. Traversal direction is also recorded because, for a reverse-traversed edge, the canonical triple is the opposite of the walk. Comparing the subject directly with the object would therefore make a valid path appear disconnected.

After these changes, all 448 multi-hop chains in the test split are genuine paths through distinct nodes. A total of 45.6% of test questions (157 of 344) have at least one such chain, rather than the 78% produced by the earlier structural label.

IX. GRAPH EMBEDDINGS

These embeddings are not evaluated as a standalone contribution. They supply the embedding-plausibility component of the graph score s_{KG} used by the dual-score re-ranker of Section X-B, which is why they are reported here at all.

TuckER, TransE, and DistMult are trained on the QA-linked graph. It has 9,803 entities, 52 relations, and 115,274 directed triples. A total of 103,746 triples are used for training and 11,528 for validation. Validation and test QA nodes are excluded from training because the graph contains answer nodes. Including those nodes could leak information from the test set.

Table XI provides a methodological warning. An earlier version of this work evaluated 200 sampled triples and reported TuckER as the best model (MRR 0.4246) and DistMult as clearly the worst (0.3273). At $n=200$, the standard error for Hits@1 is about 3.5 points, so a 0.005 difference is well within the noise. At $n=2,000$, the ordering reverses: DistMult, the simplest model, is best and TuckER is last. Model selection based on the smaller sample would therefore have been incorrect.

X. QA MODEL AND INFERENCE

The last component is the model that produces an answer. This section describes the reader, how its span scores are combined with graph scores at inference time, one decoding setting that affects the results more than any graph component does, and what the reader is actually asked.

A. Reader

The reader is InLegalBERT adapted through domain-adaptive pretraining, transfer from SQuAD v1, and IPR fine-tuning, as described in the companion manuscript [9]. The reader is kept frozen. This isolates the contribution of the graph because only the graph changes between conditions.

TABLE XII
SENSITIVITY OF VALIDATION F1 TO THE GRAPH WEIGHT, WITH $\alpha = 1 - \beta$ ($n = 408$). NO WEIGHTING IMPROVES ON IGNORING THE GRAPH.

β	0.00	0.10	0.20	0.25	0.30	0.50	1.00
F1	86.88	86.74	86.50	86.44	86.15	85.04	66.45
Changed	0	1	3	4	8	20	131

B. Dual-score inference

The reader proposes the five highest-scoring candidate spans. Each candidate receives a reader score and a graph score. The final score is

$$s(a) = \alpha s_{\text{BERT}}(a) + \beta s_{\text{KG}}(a), \quad (1)$$

with $\alpha = 0.75$ and $\beta = 0.25$. The graph score combines embedding plausibility over retrieved path chains with a cross-encoder relevance term.

C. Is the null a property of the graph or of one weighting?

Those weights were fixed before the experiments, which leaves an obvious objection: a small null could mean the graph is useless, or merely that β was set badly. “The KG does not help” and “this KG weighting does not help” are different claims, and only a sweep separates them.

Both component scores are therefore computed once per candidate span and β is swept offline, so every weighting is evaluated on identical spans and identical scores. The sweep is run on the 408 validation records, so no weighting is selected using test data, and the question is passed without the graph prefix so that $\beta=0$ is a genuinely graph-free reader.

Table XII answers it. F1 is highest at $\beta=0$ and falls monotonically as the graph is given more weight, from 86.88 to 66.45. The reported setting of 0.25 costs 0.44 F1 relative to ignoring the graph entirely, and no value in the grid recovers it. The null reported throughout this paper is therefore not an artifact of the weighting: there is no weighting of this graph score, under this scorer, that helps this reader. The α and β used in the rest of the paper are kept at their original values so that the reported system is the one that was actually built, and because moving them toward $\beta=0$ would only strengthen the negative result.

D. The effect of answer length

The decoder originally limited answers to 50 word-pieces. Answers in this benchmark average 39.9 words, so many gold spans were longer than the limit and could not be produced. This is a configuration issue rather than a modelling insight, but it is worth reporting because it is large enough to change a conclusion: in the sweep’s graph-augmented condition, adding graph text appears to improve F1 by +1.34 at a cap of 50 and to reduce it by −1.97 at a cap of 150. That condition is not identical to the question prefix measured in Section X-E, which costs 1.49 F1; the two differ in how the graph text is attached, and both are negative once the cap is large enough to emit the gold span.

TABLE XIII

THE MAIN COMPARISON AT BOTH DECODER CAPS ($n = 344$). CAP 200, SELECTED ON VALIDATION, IS THE CONFIGURATION REPORTED THROUGHOUT THIS PAPER; CAP 150 IS THE ORIGINAL TEST-SELECTED SETTING, SHOWN FOR COMPARISON. EVERY CONCLUSION IS THE SAME AT BOTH.

Condition		Cap 150		Cap 200	
		EM	F1	EM	F1
A	Clean reader	48.84	85.10	49.13	84.88
B	Prefix only	48.84	83.45	49.42	83.39
C	Re-ranking only	48.55	84.67	48.84	84.45
D	Full system	48.55	83.45	49.13	83.39
Prefix effect (B–A)		–1.65		–1.49	
Re-rank effect (C–A)		–0.43		–0.43	

A cap that can flip a sign must not be chosen on the test split. The first version of this work swept the caps $\{50, 100, 150, 200, 250, 300, 384\}$ over the 344 test records and took the value that maximised test F1, which is test-set configuration selection and turns the test split into a development split. The sweep was therefore repeated on the 408 validation records. Validation F1 rises from 78.79 at a cap of 50 to 86.88 at 150, peaks at 87.06 at 200, and is flat thereafter; the validation-selected cap is 200, not the 150 the test sweep had chosen.

The original experiments were run at 150, so the cap they used was selected on test. Rather than leave that unresolved, every experiment whose conclusion could depend on it was repeated in full at the validation-selected cap of 200: the main comparison below, the ablation ladder of Section XI-J, the conformance contrast of Section VI-C, the reader-strength ladder of Section XI-C, the graph neural re-rankers of Section XI-D1, and the passage-withheld evaluation of Section XII. No conclusion changes at the larger cap: every sign, every ordering, and every determination of significance is the same. All primary QA results in this paper therefore use the decoder cap of 200 selected on validation. The cap-150 results are retained only to document the original implementation and to show that the substantive conclusions are unchanged.

Table XIII shows that nothing in the argument turns on the choice. The clean reader is best at both caps, both graph routes lower F1 at both, the re-ranking effect is identical at -0.43 F1, and the prefix effect moves only from -1.65 to -1.49 . The absolute level is slightly lower at 200 on test and slightly higher on validation, which is what a cap selected on the other split should look like.

It is worth being precise about how far the original selection reached, since the cap-150 values appear throughout the earlier version of this work and in Table XIII. What was selected on test was the value maximising the reader’s absolute F1. Almost nothing in this paper is an absolute F1: the conformance audit does not involve the decoder at all, and every QA claim is a difference between two conditions decoded identically — prefix against no prefix, re-ranking against none, one graph against another, one reader against another. A cap chosen to raise the level of a condition raises it in both arms of such a comparison, so the difference is insulated in a way the level is not. Table XIII is the empirical check on that reasoning, and it

TABLE XIV

THE READER AT THE VALIDATION-SELECTED CAP OF 200, VARYING ONLY WHAT IT IS ASKED AND HOW THE SELECTED SPAN IS TURNED BACK INTO TEXT. ROWS ONE AND THREE REPRODUCE THE SWEEP AND TABLE XV RESPECTIVELY.

Question	Span text	EM	F1
bare q	word-pieces	49.13	84.88
bare q	source offsets	67.73	86.84
[KG: . . .] q	word-pieces	49.42	83.39
[KG: . . .] q	source offsets	65.41	85.20

behaves as the reasoning predicts: the difference that the cap should not move does not move at all, and the one that shifts does so by 0.16 F1, well inside the bootstrap width reported for effects of comparable size in Table XXI.

The analyses derived from these predictions were also recomputed at the validation-selected cap, since they are functions of the per-record output rather than of new inference. They reproduce exactly: the overall difference is $+0.00$ F1 with interval $[-0.65, +0.59]$ at both caps, the multi-hop and single-hop splits give -0.41 and $+0.35$ against -0.42 and $+0.35$, the two non-zero per-Act effects are $+0.88$ for Trade Marks and -1.51 for Geographical Indications at both, and the same four predictions change. The component ablation (Section XI-J) and the conformance experiment (Section VI-C) were re-run outright at the validation-selected cap and are likewise unchanged in ordering and sign. Every test-split analysis whose substantive conclusion could depend on the decoder cap has therefore been checked at both caps, and none changes its conclusion.

Two consequences are stated rather than left implicit. Absolute F1 values in this paper are those at the validation-selected cap of 200, and the cap-150 values are retained only as a sensitivity comparison with the original implementation; either should be compared with other work only with the decoder setting in mind. The differences, which carry every claim, should not be read that way.

E. What the reader is actually asked

The sweep above reports 84.88 F1 for the reader, while Table XV reports 83.39 F1 for the same checkpoint on the same split. The gap is not a metric difference. The inference code prepends up to two retrieved facts to the question, as [KG: . . .] q , and it does so before the reader is called. That prefix is present in every condition, including the row labelled reader only. It reaches 321 of the 344 test records, because entity linking covers the split almost completely.

Table XIV separates the two factors by running the reader once per record and varying only what happens afterwards. The bare question reproduces the sweep exactly at all seven caps, and the prefixed question reproduces Table XV exactly. The prefix therefore accounts for the whole 1.49 F1 gap.

Two consequences follow, and both matter for how the rest of this paper is read. First, the reader used here is trained on plain questions by the companion manuscript [9], so the prefix is a train–test mismatch, and injecting graph text into the question costs 1.49 F1 against a reader that never sees

it. This is a third way in which the graph fails to help, alongside re-ranking and the graph neural models, and it has the largest negative point estimate of the three. Second, every comparison below is therefore a comparison between two prefixed conditions unless stated otherwise: it isolates what re-ranking adds on top of the prefix, not what the graph as a whole is worth. The ablation in Section XI-J inherits the same property, because the prefix is built from the retrieved context after the ablation switches have been applied.

The second row of Table XIV is a separate observation, reported because it bears on how EM should be read. Reconstructing the span from the source text rather than from its word-pieces leaves F1 almost unchanged but moves EM from 49.13 to 67.73. Detokenisation reinstates statutory punctuation in the wrong places, which rarely costs a token but almost always costs an exact match. The EM values in this paper are consequently a property of the decoder as much as of the reader, which is a further reason to read F1 rather than EM on this benchmark.

XI. RESULTS

This section is organised around the research questions rather than around the order in which the experiments were run. Three are answered elsewhere: RQ1, whether the graph instantiates the ontology published with it, in Sections V and V-D, where only 19.28% of its semantic claims satisfy the declared argument types; RQ4, whether a more conformant graph changes the answers, in Section VI-C, where no effect is detectable at the resolution this benchmark allows; and RQ5, whether the graph helps once the supporting passage is withheld altogether, in Section XII, which requires a different evaluation rather than a different reading of these runs.

The two answered here are RQ2, what the graph is worth against a baseline that does not already contain it (Section XI-A), and RQ3, how that value moves with the strength of the reader and how much of it is an artifact of how the comparison is set up (Sections XI-C to XI-J). Because Section III-D has already established that this benchmark leaves no room for the knowledge-supplying benefits a legal KG is usually built for, the interest lies less in the sign of these results than in what it takes to measure the sign correctly.

All numbers come from fresh runs rather than stored scores.

A. RQ2: what the graph is worth against a clean baseline

The graph reaches the answer by two independent routes: as text prepended to the question, and as a re-ranking score over candidate spans. Reporting only the system with both routes on, against a baseline with both on but the re-ranker disabled, measures neither. Table XV therefore crosses the two, giving the four conditions that identify each effect and their interaction. All four use the same checkpoint, decoder and test split.

Three things follow. First, both routes lower F1 in isolation: the prefix costs 1.49 F1 (B against A) and re-ranking costs 0.43 F1 (C against A). Neither interval excludes zero under a paired bootstrap over the 344 records, $[-3.10, +0.03]$ and $[-1.11, +0.00]$ respectively, so what is established is that

TABLE XV
THE TWO GRAPH ROUTES CROSSED, ON THE INDICPR-QA TEST SET ($n = 344$). CONDITION D IS THE SYSTEM AS PREVIOUSLY REPORTED AND B IS THE BASELINE IT WAS PREVIOUSLY COMPARED AGAINST. A IS THE READER WITH NO GRAPH INPUT OF ANY KIND.

	Condition	Prefix	Re-rank	EM	F1	Char-F1
A	Clean reader	—	—	49.13	84.88	89.22
B	Prefix only	✓	—	49.42	83.39	86.58
C	Re-ranking only	—	✓	48.84	84.45	88.73
D	Full system	✓	✓	49.13	83.39	86.63

neither route helps, not that either is reliably harmful. The best system on this benchmark is still the one that does not use the graph at all.

Second, the previously reported comparison, D against B, gives exactly 0.00 F1. That number is valid as the incremental effect of re-ranking on top of the graph prefix, but misleading if read as the graph’s total contribution. Re-ranking is not necessarily inert; its point estimate is negative when measured against the graph-free reader, although the paired interval includes zero, and it appears inert only when measured on top of a prefix that has already moved the prediction.

Third, the two routes are sub-additive. Their combined cost is 1.49 F1 rather than the 1.92 F1 that adding the separate effects would predict, an interaction of +0.43 F1. Re-ranking costs nothing once the prefix is present ($D - B = 0.00$) and costs 0.43 F1 when it is not ($C - A = -0.43$). The natural reading is that the two routes are not supplying independent evidence: both push the prediction toward the same graph-derived spans, so the second one to be applied finds little left to change. This is consistent with the retrieved evidence being drawn from one defective layer, measured in Section XI-D1.

Conditions B and D are the comparison conventionally reported for this system. Against B, re-ranking changes EM by -0.29 points and Char-F1 by $+0.06$ points, while F1 is unchanged. The per-Act and significance analyses that follow all use this D-against-B contrast, so they quantify re-ranking rather than the graph as a whole.

Only 4 of 344 predictions change. Three predictions improve, while one changes substantially for the worse. A Geographical Indications question loses 76.9 F1 because the graph promotes a plausible but incorrect span. By Act, the graph improves Trade Marks by $+0.88$ F1, reduces Geographical Indications by -1.51 F1, and has no effect on the other Acts.

B. Multi-hop questions

The graph was expected to help when the retrieved evidence had multi-hop structure. Under the corrected chain definition, 157 of 344 test questions qualify. The graph gives -0.42 F1 on these questions and $+0.35$ F1 on the remaining 187 questions. Neither result is significant. The bootstrap 95% intervals are $[-1.47, +0.15]$ and $[0.00, +1.05]$, and the overall interval is $[-0.65, +0.59]$.

The direction here is opposite to what a permissive chain definition and a weaker reader would suggest, and neither result differs reliably from zero. Multi-hop structure in the retrieved subgraph does not identify questions the graph helps.

TABLE XVI

WHAT THE GRAPH IS WORTH AGAINST READER STRENGTH ($n = 752$). EVERYTHING EXCEPT THE READER IS HELD FIXED. RE-RANK Δ IS MEASURED AGAINST THE PREFIXED CONDITION, WHICH IS THE CONVENTIONAL COMPARISON; NET IS MEASURED AGAINST THE READER GIVEN NO GRAPH INPUT AT ALL.

Reader	Clean	Prefix	+Re-rank	Re-rank Δ	Net
SQuAD-only	31.09	26.16	27.23	+1.07	-3.86
SQuAD-only (partial)	31.13	26.17	27.16	+0.99	-3.97
SQuAD-only, 2 epochs	32.61	28.14	29.09	+0.95	-3.52
Plain encoder + IPR, 1k	45.10	45.08	45.07	-0.01	-0.03
Plain encoder + IPR, 2k	60.51	56.65	56.61	-0.04	-3.90
Plain encoder + IPR, 3k	76.87	73.62	73.69	+0.07	-3.18
Plain encoder + IPR, 4k	78.86	77.11	77.03	-0.08	-1.83
Plain encoder + IPR	79.36	77.43	77.38	-0.05	-1.98
DAPT + IPR, 2 epochs	85.91	83.17	83.18	+0.01	-2.73
DAPT + IPR, 3 epochs	86.06	83.44	83.46	+0.02	-2.60

C. RQ3: how the value moves with reader strength

The results above show that a KG gain can disappear when the baseline changes, and Section X-E showed that it can also be manufactured by a baseline the graph has already degraded. To test how both behave across reader strength, the same pipeline was applied to ten distinct reader checkpoints from the training process. The graph, subgraphs, scorers, α , β , and decoder were unchanged.

This curve is evaluated on the validation and test splits together, for a total of 752 records, rather than on the test split alone. The reader-strength curve is one of the main empirical analyses of this paper, and 344 records is a small sample. Pooling is appropriate here because no model or parameter is selected using these records. Every reader is a fixed checkpoint from an earlier training run, and the KG pipeline has no free parameter fitted on these records. The test-only curve has the same shape.

Each reader is evaluated in three conditions rather than two, because Section X-E showed that the condition previously called the baseline already carries graph input. The three are the reader alone, the reader with the [KG:...] question prefix, and the full system with re-ranking on top of the prefix. The prefix column is recomputed here rather than reused, and it reproduces the previously reported base column for every checkpoint, which confirms that the two harnesses agree. Table XVI gives the three conditions per reader and Figure 3 plots them, with the net effect of the graph against the graph-free reader in the lower panel.

Checkpoint identity was checked by weight hash before anything was computed. An earlier version of this ladder listed the final DAPT reader and its third-epoch checkpoint as separate readers; the two files are byte-identical, and a second such pair existed in the plain-encoder run. Both duplicates are removed, so the ten rows are ten distinct models spanning eight F1 levels from 31.01 to 86.07. This matters because the correlation has one observation per reader, not one per record.

Read along the re-rank column, the pattern is consistent and is reported as a description of this ladder rather than as an estimate for readers in general, for the reason given below. The benefit of re-ranking falls from +1.07 F1 at the weakest reader to +0.02 at the strongest, and it is essentially spent by

TABLE XVII

CORRELATIONS ACROSS THE READER LADDER. THE UNIT OF OBSERVATION IS THE CHECKPOINT, NOT THE QA RECORD: EACH READER CONTRIBUTES ONE POINT, SCORED ON 752 RECORDS. INTERVALS ARE PERCENTILE BOOTSTRAP OVER READERS, 10,000 RESAMPLES.

	Re-rank Δ vs prefixed base	Net KG vs clean base
Readers n	10	10
Pearson r	-0.865	+0.294
95% CI	[-0.99, -0.06]	[-0.36, +0.90]
Spearman ρ	-0.600	+0.552
95% CI	[-0.99, +0.22]	[-0.18, +0.92]

45 F1: every reader above that point lies within ± 0.15 F1 of zero. The count of changed predictions falls from 40 to 12; at the weak end most changes are corrections (33 improved, 7 worsened), at the strong end they are nearly even (7 and 5).

Read along the net column, the picture is different and the earlier version of this paper overstated it. On five checkpoints the net effect was negative everywhere and appeared to shrink monotonically with reader strength. With ten it does neither. All ten net point estimates are negative, ranging from -3.97 to -0.03 F1, but the relationship with reader strength is not reliably monotonic: the net-effect correlation is +0.294 and its interval spans zero (Table XVII). These values are descriptive. Because per-record intervals were not computed for the clean arm, they should not be read as evidence that the graph is significantly harmful at every reader strength, only that its net point estimate is below zero at each one.

Table XVII separates the two relationships, and they behave in opposite ways once the ladder is filled in. The re-ranking effect survives: the interval excludes zero under both coefficients, and the point estimate falls from the -0.983 obtained on five clustered checkpoints to -0.865, which is the more honest number. The net effect does not survive: at +0.294 with an interval spanning zero it is not supported at all, and the +0.960 obtained on five checkpoints was an artifact of three tight clusters and a duplicated row.

The ladder was also run at the original cap of 150 (Section X-D), all ten readers and both arms. The re-ranking correlation is -0.898 there with a 95% interval of [-0.99, -0.67], against -0.865 and [-0.99, -0.06] here: the same sign and the same conclusion, with the interval at the validation-selected cap the wider of the two. The net-effect correlation is +0.294 at both caps, spanning zero at both. The delta at the weakest reader is +1.07 F1 here against +1.13 at the smaller cap, and +0.02 at the strongest in both, so the decay across the ladder is not an artifact of either cap.

One caveat applies to both columns. The ten checkpoints come from three training runs, so adjacent points within a run are correlated by construction and are not ten independent draws from a population of readers. The bootstrap treats them as exchangeable, which is optimistic, and the intervals should be read as descriptive of this ladder rather than as inference about readers in general.

The practical implication sharpens accordingly. Reporting baseline strength is necessary but not sufficient: the baseline must also be free of the intervention being tested. Every

TABLE XVIII
GRAPH NEURAL RE-RANKERS, ALL AGAINST THE SAME READER.

System	EM	F1	Δ vs dual
Prefix only, no re-ranking	49.42	83.39	N/A
Prefix + dual-score KG	49.13	83.39	0.00
QL-RGCN	48.84	83.33	-0.06
QL-CompGCN	49.13	83.17	-0.22
QL-HGT	48.84	82.59	-0.80

number in the re-rank column of Table XVI is a defensible measurement of what re-ranking contributes, and every one of them would be read by a reviewer as what the graph contributes. Across the ladder those two readings differ by between 0.44 and 5.22 F1, and they differ in sign at seven of the ten readers.

D. Graph neural re-rankers

If the problem were the simplicity of the scoring function, a stronger graph model should help. Therefore, three models are evaluated: a relational GCN, a heterogeneous graph transformer, and a relation-composition GCN.

Table XVIII shows that all three models perform worse than the simple weighted score. The weighted score itself provides no gain. Every row here carries the question prefix of Section X-E, including the first, so the comparison is between graph re-rankers and not against a graph-free reader; the graph-free reader reaches 84.88 F1 and beats all of them. The ordering is prefix-only = dual-score > all three GNNs. This indicates that graph-model capacity is not the limiting factor.

Re-running the three re-rankers over candidate spans decoded at the original cap of 150 (Section X-D) leaves this unchanged. There the dual score again matches the prefix-only reader exactly, at 83.45 F1, and the three graph models again fall below it, by -0.34 (QL-RGCN), -0.41 (QL-CompGCN) and -0.73 (QL-HGT) F1. Both the ordering and the sign of every entry are the same at both caps.

1) Why the graph neural models fail: A possible explanation is that message passing spreads noise rather than legal evidence. This was tested by measuring the edges used by the graph re-ranker. For every test question, the retrieved subgraph was taken and each edge was checked against the ontology.

Of 1,751 retrieved edges, only 128 (7.31%) are conformant. This is lower than the 19.28% conformance of the full graph. The reason is structural: retrieval filters out mentionedIn and the other structural relations, which are 100% conformant. The retrieved subgraphs therefore contain mainly the semantic layer, which is the defective layer. Every retrieved edge in the test set asserts a legal fact, and 92.7% of these edges fail the ontology-conformance test. Whether they are also legally false is a separate question this measurement does not answer; see Section VI-A.

Three of the six most frequently retrieved relations are conformant 0.00% of the time: revokedBy (331 occurrences), definesAs (242), and imposesObligation (218). The other three are barely better: grantedBy (309) is at 1.29%, appliesTo (216) at 0.46%, and only hasTermOf (222) reaches 31.53%. A GNN

TABLE XIX
LOCAL LLM BASELINES FROM THE PROMPTING HARNESS VERSUS THE ADAPTED READER.

System	ZS EM	ZS F1	FS EM	FS F1
Mistral-7B	2.62	55.55	24.13	67.15
Gemma3-12B	34.30	67.48	37.21	70.38
Llama3.1-8B	24.71	68.25	33.43	72.51
Qwen2.5-7B	38.08	75.16	34.30	74.78
Reader + KG (this work)	49.13 EM / 83.39 F1			

that aggregates over such a neighbourhood has little schema-conformant structure to propagate.

A second measurement supports this result. If retrieved evidence were useful, questions with more conformant evidence should receive better answers. The Pearson correlation between retrieved-edge conformance and F1 is $r = 0.021$, which is effectively zero. Section VI-C strengthens this observation with a controlled contrast and reaches the same conclusion.

E. LLM baselines

Table XIX reports local LLM baselines on the same test set. The adapted encoder reader outperforms all of them at roughly 110M parameters: 64 times smaller than the 7B baselines and 109 times smaller than Gemma3-12B.

There is a discrepancy with the companion DAPT manuscript [9]. The manuscript reports higher zero-shot values for the same models on the same split. For example, it reports 67.11 F1 for Mistral-7B, compared with 55.55 in this work. The difference comes from prompting and answer extraction, not from the models or the data. The harness used in this work records more refusals. Readers comparing the two studies should use the companion manuscript's harness as the reference.

F. Error analysis

Of the 344 test predictions of the reported system, 169 match the gold answer exactly, 62 more reach at least 0.9 F1, 63 overlap it between 0.5 and 0.9, 44 between 0.1 and 0.5, and 6 are effectively disjoint. Most non-exact predictions are therefore close to the gold answer rather than wrong: 231 of 344 are within 0.1 F1 of it. This is why token and character F1 are more informative than EM for this dataset, and it is the same distribution that Section III-D uses to argue that most of the residual error is a span-boundary problem rather than a missing-knowledge problem.

This classification describes the symptoms. To find the causes, the 30 lowest-scoring predictions were assigned to the stage that failed. Each example used its recorded pipeline state: whether entity linking returned an entity, whether retrieval produced a path, whether any retrieved edge was conformant, and whether the gold span was present in the supplied context.

In 60% of the worst cases, the graph supplied evidence, but every piece of evidence was non-conformant. Only 10% were retrieval failures in the usual sense of finding no evidence. The main failure is evidence that does not satisfy the schema rather than evidence that is missing. This is the same conclusion reached by the conformance analysis.

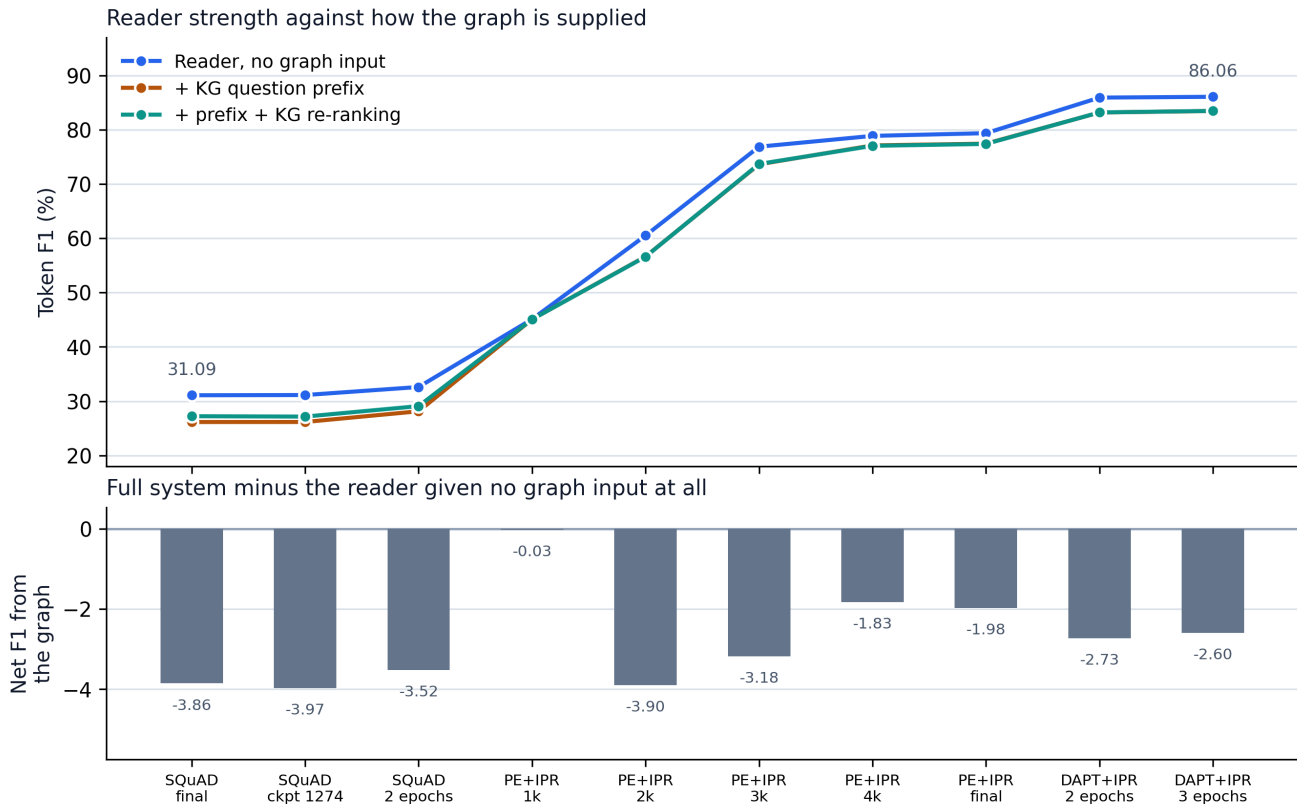


Fig. 3. Three conditions per reader across ten distinct checkpoints, and the net effect of the graph against a reader given no graph input. The net point estimate is negative at all ten checkpoints. Each point is scored on 752 held-out records; see Table XVII for the correlations and their intervals.

TABLE XX
ATTRIBUTED CAUSE FOR THE 30 WORST PREDICTIONS.

Cause	Count	Share
All retrieved edges non-conformant	18	60%
Reader error despite usable evidence	9	30%
Retrieval returned no path	3	10%

G. Two worked examples

Aggregate numbers do not show how the graph changes individual answers. The largest gain and the largest loss are therefore traced below.

Where the graph helps. For `tm_000870` (Trade Marks, +65.3 F1), the question asks what clause (b) of Section 123 lays down. The reader stops early and returns “the defendant raises a defence under clause”. Re-ranking promotes a candidate that contains the full statutory clause. This candidate closely matches the gold span. The graph therefore selects the correct answer.

However, it does not select the answer for the correct reason. The chain supporting the promoted span is `person for the time` → `definesAs` → `person`. This is not a meaningful legal fact and is unrelated to Section 123. The span was promoted because a retrieved node string overlapped with it, not because the graph encoded a supporting statutory relationship.

Where the graph hurts. For `gi_000360` (Geographical Indications, -76.9 F1), the question asks what Section 76

provides regarding agents. The reader returns “a legal practitioner, or”, which exactly matches the gold span. Re-ranking replaces it with a much longer span that continues through clauses (b) and (c), causing most of the F1 loss.

The supporting chain is `register` → `definesAs` → `register` of patents referred to in section 67. A Patents definition was retrieved as evidence for a Geographical Indications question. The edge itself comes from definition repair, and the contamination comes from the node it hangs on: entity nodes are keyed by surface form and are not scoped to an Act, so the single node `register` carries every Act’s statements about a register at once. A Geographical Indications question that mentions the register therefore reaches a Patents definition in one hop. Nothing in the graph records that the hop crossed a statute. It changes a correct answer into an incorrect one.

Together, these cases explain the aggregate zero. The graph changes very few answers. When it changes an answer, it may be correct for the wrong reason or incorrect altogether.

H. Statistical significance

Because the effects are small, percentile bootstrap confidence intervals are reported for every comparison, not only for the multi-hop split. The analysis uses 10,000 resamples and seed 42. The intervals are computed over paired per-question differences.

Every interval in Table XXI contains zero. The two largest per-Act effects are therefore not distinguishable from noise.

TABLE XXI
BOOTSTRAP 95% CONFIDENCE INTERVALS FOR KG RE-RANKING
AGAINST THE READER. NO COMPARISON IS SIGNIFICANT.

Comparison	<i>n</i>	Δ F1	95% CI
All questions (F1)	344	+0.00	[−0.65, +0.59]
All questions (EM)	344	−0.29	[−0.87, +0.00]
Multi-hop questions	157	−0.42	[−1.47, +0.15]
Single-hop questions	187	+0.35	[+0.00, +1.05]
Trade Marks	88	+0.88	[+0.00, +2.49]
Geographical Indications	51	−1.51	[−4.52, +0.00]
Other five Acts	205	+0.00	[+0.00, +0.00]

Trade Marks has a gain of +0.88 with interval [+0.00, +2.49], and Geographical Indications has a loss of −1.51 with interval [−4.52, +0.00]. These values are reported because they are the only non-zero per-Act effects, not because they are established effects.

I. Variation across Acts

These are the only two Acts where re-ranking changes predictions, and the effects have opposite signs. It is therefore useful to ask what distinguishes them. The obvious hypothesis is graph quality: the graph should help where its edges are more accurate.

The data do not support this hypothesis. Geographical Indications has the highest retrieved-edge conformance of any Act (10.51%), yet it is the Act that the graph harms most (−1.51 F1). Trade Marks has the lowest conformance (5.41%), yet it is the only Act that benefits (+0.88 F1). Retrieval volume does not explain the difference either. GI retrieves more entities per question (4.27 versus 2.92) and slightly more chains (5.41 versus 5.25).

This work does not claim to have identified a mechanism. Together with the near-zero correlation between evidence conformance and F1 ($r = 0.021$), the results suggest that graph quality does not determine the sign of the per-Act effect. Only four records change across the full test set: three in Trade Marks and one in GI. Differences based on so few records should not be interpreted as general Act-level behaviour.

J. RQ3: which component carries the effect

Each variant in Table XXII disables one component and re-runs the full inference path. The distinction matters: reconstructing a variant from stored candidate scores yields identical numbers for components that were never disabled, and so cannot distinguish “this component does nothing” from “this component was still running”.

The result is clear: removing graph components improves the system. Removing the learned embeddings, the multi-hop chains, or the entire entity-linking and retrieval stage increases F1 by +0.22 and EM by +0.29. Disabling the KG score or the cross-encoder does not change the result.

No component produces a positive effect. The two apparently neutral components are neutral because the graph score they provide is already ignored in most cases. The three harmful components are the ones that affect ranking. The best

TABLE XXII
EXACT COMPONENT ABLATION. EACH ROW IS A FULL RE-RUN OF
INFERENCE WITH ONE COMPONENT DISABLED, NOT A RECONSTRUCTION
FROM STORED SCORES.

Variant	EM	F1	Δ F1
Full system	49.13	83.39	N/A
− KG score ($\beta=0$)	49.42	83.39	0.00
− cross-encoder	49.13	83.39	0.00
− KG embeddings (Tucker)	49.42	83.61	+0.22
− multi-hop path chains	49.42	83.61	+0.22
− entity linking + retrieval	49.42	83.61	+0.22

configuration of the system is the configuration with the graph switched off.

One component was originally unreachable from this table. The question prefix of Section X-E is rebuilt from the retrieved context after the ablation flags are applied, so it survived every row, including the row that disables entity linking and retrieval; that row reaches 83.61 F1 rather than the 84.88 F1 of a reader given no graph input at all. The released code now carries two further switches, `no_prefix` and `no_graph`, and the latter is the bottom of the ladder: it reproduces 84.88 F1, matching the independently measured clean reader of Table XV exactly.

The ladder therefore reads as a single monotone sequence. Removing the graph from the ranker is worth +0.22 F1; removing it from the question as well is worth a further +1.27; and the configuration with no graph anywhere is the best of all. No ordering of these components recovers a positive contribution.

Every row was also computed at the original cap of 150 (Section X-D). The ladder is unchanged: the two components that do nothing still do nothing exactly (0.00 F1 at both caps), the three that affect ranking are worth +0.23 F1 there rather than +0.22, and removing all graph input is worth +1.65 F1 rather than +1.49. The ordering, the sign of every entry, and the conclusion that the best configuration is the one without the graph are identical at both caps.

XII. PASSAGE-WITHHELD RETRIEVAL EVALUATION

Every result reported so far hands the reader the statutory passage that contains the answer, which is what bounds the QA claim to extractive benchmarks of this shape (Section III-D). This section removes the passage and measures what happens.

RQ5. Does the knowledge graph help when the relevant statutory passage is not supplied to the reader?

The system now sees the question alone and must rank all 1,116 sections of the seven Acts to find the provision the question came from. Two conditions were planned. In R1 — passage-withheld, provision-named retrieval — the passage is withheld and the question is left unchanged, so it usually still names the provision. In R2 the explicit section reference would also be masked, so that the retriever has to identify the provision rather than be told it. Section XII-A reports that R2 cannot be constructed from this benchmark and why; Sections XII-B to XII-D report R1, first as retrieval and then as end-to-end QA against the passage-supplied oracle.

TABLE XXIII

CONSTRUCTIBILITY OF A SECTION-BLINDED RETRIEVAL CONDITION. EACH QUESTION IS CLASSIFIED BY WHAT INFORMATION SURVIVES REMOVAL OF ITS STATUTORY SECTION REFERENCE. ONLY THE LAST ROW WOULD SUPPORT A GENUINE RETRIEVAL-REQUIRED QUESTION; IN THE FIRST ROW THE SECTION REFERENCE IS THE ENTIRE QUESTION.

What survives blinding	Train	Val	Test
Pointer only	2,289 (61.7%)	227 (55.6%)	202 (58.7%)
Copied from title	875 (23.6%)	99 (24.3%)	105 (30.5%)
Copied from passage	456 (12.3%)	71 (17.4%)	37 (10.8%)
Usable when blinded	90 (2.4%)	11 (2.7%)	0 (0.0%)
Total	3,710	408	344

A. Whether a section-blinded condition can be built

R2 is the condition that would matter most, because Table I records that 94.48% of test questions name their own statutory section. If the question already says which provision to read, retrieval is close to a lookup. The obvious repair is to delete the section reference and keep the rest of the question.

That repair does not survive contact with the data. Each question was classified by what information remains once its section reference is removed, separating content that is genuinely the reader’s question from content copied out of the material the retriever is supposed to find. The classification is deterministic code rather than manual judgement, so that the claim can be rerun and checked. Statutory identifiers are stripped with a regular expression covering the forms the generator produced (Section 14, sec. 9, Section 30A, Section 18(3)), as are bare sub-references such as clause (b); the remaining tokens are filtered against a fixed list of template scaffolding, the words contributed by the question generator rather than by the question. A question is pointer only when fewer than two content words survive that filter. Of those that survive, it is copied from section title when fewer than two of its surviving words of more than three characters are absent from the section heading, and copied from gold passage when at least 60% of those remaining words also occur in the answer span or its surrounding context. Anything else is usable. The thresholds are deliberately generous to the usable class, so the count in that row is an upper bound. Table XXIII gives the result, and the released audit file lists every question with its category and its mechanically blinded form, with empty columns for a human reviewer’s verdict and rewrite.

In 58.72% of test questions nothing remains. These are questions such as “What does sub-section (1) of Section 15 provide?”, where the section reference is not a hint attached to a question but the entire question; removing it leaves a string that identifies no provision. A further 30.52% survive only because the question quotes the section title verbatim, which is an exact lexical match against the gold document and therefore replaces one retrieval hint with another. In 10.76% the surviving content was lifted from the gold passage itself, as in “Under Section 77, what shall any other matter which be?”, where the question already contains the text it asks about. No test question falls outside these three classes. Blinding also destroys the identity of the queries: the 344 test questions collapse to 233 distinct strings, so questions with different gold

sections become indistinguishable inputs that no retriever can separate.

The conclusion is not that the masking was implemented badly. The questions were generated from section numbers and titles, so for most records the section reference carries the entire retrieval signal, and a benchmark of this construction cannot express the task R2 was meant to measure. This sharpens the headroom argument of Section III-D rather than repeating it: that the question usually names its section invites the reply that the naming could simply be removed, and the measurement here is that for most test questions there is nothing left to ask once it is. Constructing R2 would require re-authoring the questions from section content, which is new data collection rather than a transformation of the existing benchmark, and is left to future work (Section XV).

R1 is therefore a diagnostic: the questions retain their section references, so the numbers below describe an easy retrieval setting and are not evidence about retrieval when the provision is genuinely unknown.

B. Retrieval systems

The corpus is the 1,116 distinct statutory sections from which IndicIPR-QA was built, each indexed as one document by heading and body. The gold document for a question is the section its supplied context came from, recovered by exact text match, so the labels require no annotation and are unambiguous for all 4,462 records. Four systems rank that corpus.

BM25 [39] is the lexical baseline. Two dense encoders are included, because a graph compared only against a lexical retriever proves little, and because the two answer different questions. InLegalBERT [7] is the in-domain encoder this pipeline already uses for entity linking (Section VIII); as a masked-LM checkpoint never trained with a contrastive retrieval objective, it shows what legal-domain pretraining alone provides. BGE-small-en-v1.5 [40] is retrieval-trained and is the competent dense baseline. IPRon retrieval links the question to graph entities, expands over the legal graph, and scores sections by the occurrence of the activated entities. The hybrid fuses all four by per-query min-max normalisation.

The leakage control matters more here than the systems do. The QA links of Section VII — `has_answer`, `from_section` and `supported_by` — exist exactly once per benchmark record and encode which section holds the answer, so using them would let the graph read the label. This experiment therefore loads the legal graph only, the loader aborts if any benchmark-derived relation is present, and the subgraph fields precomputed in Section VIII are not consulted, having been built with the gold context in scope. The only path from a question to a section is entity linking followed by graph expansion.

Every hyperparameter was selected on validation by MRR and then frozen, and the test split was scored once, following the correction of Section X-D. The selected values are $k_1 = 0.9$ and $b = 0.6$ for BM25; 256 tokens for InLegalBERT and 512 for BGE-small, both ℓ_2 -normalised under cosine similarity, mean-pooled and CLS-pooled respectively, with the

TABLE XXIV

RETRIEVAL-REQUIRED EVALUATION (CONDITION R1). THE GOLD PASSAGE IS WITHHELD AND EACH SYSTEM RANKS ALL 1,116 STATUTORY SECTIONS. HYPERPARAMETERS WERE SELECTED ON VALIDATION AND FROZEN; TEST WAS SCORED ONCE. THE QUESTIONS STILL NAME THEIR OWN SECTION, SO THIS IS AN EASY RETRIEVAL SETTING.

System	R@1	R@5	R@10	MRR
BM25	34.9	60.5	73.3	0.467
Dense (InLegalBERT)	10.8	15.1	17.7	0.137
Dense (BGE-small)	35.5	51.2	57.0	0.433
IPROn KG	0.3	1.7	2.6	0.015
Hybrid (tuned fusion)	42.4	68.9	76.2	0.545
Hybrid, no graph	42.1	69.5	77.3	0.542
Random ranking	0.09	0.45	0.90	0.0068

recommended query instruction for BGE-small; and five entity links, two hops and a decay of 0.5 for the graph. Fusion weights were searched over the four-component simplex in steps of 0.1, and validation did not distribute them evenly: the graph received 0.1, BM25 0.4, BGE-small 0.5, and InLegalBERT exactly 0.0 — the grid was free to use it and declined. The fused system is therefore two text retrievers plus the graph, even though four signals entered the search, and its no-graph variant is BM25 and BGE-small at 0.5 each. All grids, seeds and per-query rankings are released.

C. Retrieval results

Table XXIV reports the frozen systems on the test split, scored by recall at 1, 5 and 10 and by mean reciprocal rank [41].

Under the retrieval procedure evaluated here, IPROn performs very poorly as a statutory-section retriever. It ranks the correct section first for one question out of 344 and within the top ten for nine, against a random-ranking floor of 0.09% and 0.90%. The performance is two to four times chance and provides little usable retrieval signal in practical terms. What is tested is one procedure — entity linking, hop expansion, and scoring sections by the occurrence of activated entities — and not the graph’s suitability for every possible retrieval algorithm, so the finding is about this pipeline rather than about IPROn in principle. The result is nevertheless consistent with the earlier conformance audit: the semantic layer is 19.28% conformant and two thirds of its failures involve an endpoint the pipeline never assigned a specific class (Section V), which is the entity grounding that a procedure of this kind depends on.

The tuned fusion is the strongest system at 68.9 R@5, and validation assigned the graph a non-zero weight of 0.1. That weight is not evidence that the graph contributes, because a small weight fitted on validation can be fitting noise, so the fusion was re-tuned with the graph removed entirely and scored on test once. Differences are reported with paired percentile bootstrap intervals over per-record scores, 10,000 resamples, as in Section XI-H. Including the graph is worth +0.29 R@1 (95% CI $[-1.16, +1.74]$), -0.58 R@5 ($[-2.33, +1.16]$), -1.16 R@10 ($[-2.91, +0.29]$) and +0.002 MRR ($[-0.01, +0.01]$). Every interval spans zero, so the graph’s contribution to the fused retriever is not distinguishable

from nothing, and its point estimate is negative at the two cutoffs that matter for feeding a reader. The advantage of the hybrid over BM25 — +7.56 R@1 ($[+3.78, +11.63]$) and +8.43 R@5 ($[+4.07, +12.79]$), both excluding zero — therefore comes from combining two text retrievers, not from the graph. That advantage is itself concentrated at the sharp cutoffs: at R@10 the hybrid gains +2.91 with an interval of $[-1.16, +6.98]$ that spans zero.

The two dense encoders separate cleanly, and the separation is itself a result. On identical inputs, identical pooling code and the same frozen protocol, the retrieval-trained encoder reaches 0.433 MRR while the legal-domain masked-LM checkpoint reaches 0.137, below BM25 by a wide margin. In-domain pretraining does not by itself produce a usable retriever, which is worth stating because InLegalBERT is a common default in Indian legal NLP and is used for entity linking in this very pipeline.

The scope of R1 is correspondingly narrow. Because the questions retain their section references, BM25 at 60.5 R@5 reflects the ease of matching a question that names its own provision, not the difficulty of statutory retrieval in general. What the condition supports is therefore limited: when the passage is withheld but the section is still named, ordinary text retrieval solves most of the task and the graph adds nothing to it. Whether a graph would help when the provision is genuinely unnamed is the question R2 was designed to ask, and Section XII-A shows this benchmark cannot ask it.

D. Downstream QA

Retrieval metrics answer RQ5 for the graph, but they do not say what the withheld passage was worth to the system as a whole. That question is answered in the unit the rest of this paper is written in. The retrieved sections were given to the same reader used in Section XI-A, at the same decoding cap, scored with the same functions; the only change is what occupies the context slot.

The multi-section conditions need their input construction stated exactly. For top- k , the k retrieved sections are concatenated in ranking order, each as its heading followed by its body and separated by a blank line, and the result is passed to the reader as a single context. The reader tokenises it to a maximum of 384 word-pieces with a stride of 128 and overflow enabled, so a concatenation longer than one window is split into overlapping windows rather than truncated; candidate spans are extracted from every window, deduplicated by surface form with the best score retained, and the highest-scoring span over all windows is the prediction. The reader is therefore run once per question over the whole concatenation, not once per section. Nothing about the gold section is used when ordering, concatenating or windowing the retrieved passages.

Table XXV reports the result. The first row supplies the passage the question was written from and reproduces the graph-free condition of Section XI-A exactly, at 49.13 EM and 84.88 F1, which fixes the oracle and confirms that the rows below are comparable to the numbers already reported.

Replacing the gold passage with the top-5 evidence returned by the best retriever reduces F1 from 84.88 to 43.86, a decrease

TABLE XXV

DOWNSTREAM QA WHEN THE PASSAGE MUST BE RETRIEVED. THE FIRST ROW IS THE ORACLE: THE READER RECEIVES THE PASSAGE THE QUESTION WAS WRITTEN FROM, WHICH IS THE CONDITION EVERY OTHER RESULT IN THIS PAPER USES. THE REMAINING ROWS REPLACE IT WITH WHAT EACH RETRIEVER RETURNED, CHANGING NOTHING ELSE. THE LAST PAIR REPEATS THE FUSED SYSTEM WITH THE GRAPH REMOVED, SO THE GRAPH’S CONTRIBUTION CAN BE READ AT THE END OF THE PIPELINE AS WELL AS AT THE RETRIEVAL STAGE. RETRIEVAL RECALL IS REPEATED FROM TABLE XXIV.

Reader input	Recall	EM	F1	Char-F1
Gold passage (oracle)	100.0	49.13	84.88	89.22
BM25 top-1	34.9	15.12	36.00	58.13
BM25 top-5	60.5	17.15	39.38	65.91
Dense (BGE-small) top-1	35.5	13.95	37.38	62.94
Dense (BGE-small) top-5	51.2	13.95	36.90	65.81
IPRON KG top-1	0.3	0.00	15.57	53.01
IPRON KG top-5	1.7	0.29	14.90	51.44
Hybrid top-1	42.4	16.86	41.80	63.66
Hybrid top-5	68.9	19.48	43.86	67.90
Hybrid, no graph, top-1	42.1	16.86	41.25	62.52
Hybrid, no graph, top-5	69.5	18.90	43.77	67.83

of 41.0 points (95% CI $[+36.8, +45.3]$), and EM from 49.13 to 19.48, a decrease of 29.65 ($[+24.71, +34.59]$). Both intervals are far from zero. This difference shows how strongly the reported QA score depends on the benchmark supplying the answer-bearing passage, and it is the operational counterpart of what Section III-D established structurally: that section showed the answer is inside the supplied context in 100% of records and inferred that the knowledge-supplying role of a graph could not be tested, while the present table shows what the same system scores when that support is withdrawn.

The IPRON top-1 condition produces 0.00 exact match. Reading from the single section IPRON ranks highest: across 344 questions the reader never once recovers the exact answer, while token F1 remains at 15.53. The combination of 0.00 EM with 15.53 token F1 shows that substantial token overlap can persist even when the retrieved section is almost never the gold section, which is expected for statutory text, where any section of an IPR Act shares drafting vocabulary with any other. Token-overlap metrics should therefore be interpreted cautiously in this domain: a system that retrieves the wrong provision still scores well above zero on F1 alone.

Two smaller observations follow from the table. Additional retrieved context helps only when retrieval is accurate enough to repay it: top-5 gains 3.43 F1 over top-1 for BM25 and 2.31 for the hybrid, whose recall rises steeply with k , but costs 0.60 F1 for the dense retriever, whose recall rises more slowly, so the extra passages add more distraction than evidence. And the downstream ordering follows the retrieval ordering without inversions, which is the expected result but worth confirming, since it establishes that nothing in the reader compensates for a weak retriever. The downstream margin of the hybrid survives the same test as the retrieval margin: +4.53 F1 over BM25 at top-5 ($[+0.93, +8.15]$) and +7.44 over the dense retriever ($[+3.52, +11.29]$).

The graph’s null contribution also survives the reader, which is the natural next question once Section XII-C has shown it at the retrieval stage. Deleting the graph from the fused

system and re-reading changes F1 by +0.47 at top-1 (95% CI $[-0.79, +1.78]$) and +0.21 at top-5 ($[-1.94, +2.37]$), and EM by 0.00 ($[-1.16, +1.16]$) and +0.58 ($[-1.45, +2.62]$). Every interval spans zero. The graph therefore does not detectably change the answers either through the ranking it produces or through the evidence the reader ends up reading, which is a stronger statement than the retrieval comparison alone supports.

The whole subsection was also computed at the original cap of 150 (Section X-D). Nothing changes. There the oracle is 48.84 EM and 85.10 F1 and the best retrieved condition 19.19 EM and 43.89 F1, so the fall from oracle to best retrieved is 41.20 F1 against 41.02 here, with an identical EM fall of 29.65. IPRON returns 0.00 EM at top-1 under both caps, and deleting the graph from the fused system moves nothing that its interval excludes under either: +0.47 F1 at top-1 and +0.21 at top-5 at the smaller cap, against +0.55 ($[-0.72, +1.86]$) and +0.09 ($[-2.08, +2.26]$) here. Both cap settings are released.

The scope of this subsection is the same as that of Section XII-C. These numbers describe the setting in which the passage is withheld but the section is still named. They do not describe legal QA with genuinely unknown provisions, which Section XII-A shows this benchmark cannot pose.

XIII. DISCUSSION

The results say that the graph does not improve answers on this benchmark. This leaves two questions: what the graph is still useful for, and why it fails at the task it was built for. The following sections address both and draw lessons for KG-augmented QA beyond Indian IPR law.

A. Where the graph helps and where it does not

On this benchmark, the graph does not improve answer accuracy. The graph might still be useful for explanation even if it does not improve accuracy. Figure 4 shows this possibility for the test question with the cleanest retrieved evidence.

This case has some positive features. Retrieval reaches the correct statutory section, and three of the five retrieved edges satisfy the conformance test: patent application $\rightarrow$ submittedTo $\rightarrow$ controller, layout-design $\rightarrow$ regulatedBy $\rightarrow$ controller, and patent $\rightarrow$ governedBy $\rightarrow$ patents act. These are readable claims that a lawyer could check, which is what conformance is a precondition for; whether each is legally accurate is a separate question this measurement does not settle, and the second is doubtful, since layout-designs are administered under their own Act rather than by the patent Controller.

The other two edges fail the test. application $\rightarrow$ grantedBy $\rightarrow$ its issue is meaningless, and one object is a truncated clause fragment. This is the best case in the test set. It contains three conformant claims and two that a user would need to discard. The ratio is worse in other cases. Section XI-G shows a Geographical Indications question supported by a Patents definition.

The graph is therefore less useful for explanation than the phrase “useful for explanation” suggests. IPRON is a structured, queryable representation of seven statutes, with every benchmark question and answer linked to the graph.

The conformance validator makes its weaknesses measurable rather than hidden. The principal reusable contribution is the audit methodology; IPROn serves as the case study through which that methodology is developed and evaluated. With `imposesObligation`, `definesAs`, `grantsRight` and `hasRemedyFor` conformant at zero and the two largest properties near it, IPROn is best understood as an audit artifact rather than as a legal knowledge base, and should not be shown to a legal user without filtering.

B. Why the graph does not help

The measurements provide three reasons, and they are not equally fundamental. The first is the benchmark. Because the answer-bearing passage is already supplied and the relevant provision is usually named (Section III-D), the headroom for knowledge-supplying and unnamed-provision retrieval benefits is close to zero before any graph is introduced. No improvement in graph quality can create headroom for those two mechanisms when the benchmark already supplies the information, which is why this paper does not present its QA result as a finding about knowledge graphs in general. Mechanisms that do not depend on missing information, such as span selection, remain available and are measured separately.

Withholding the passage does not rescue the graph, which answers the first objection the headroom argument invites. Required to find the provision themselves, IPROn ranks the correct section first for one test question in 344, shows no detectable contribution to a fused text retriever, and yields 0.00 exact match downstream (Section XII). The benchmark limitation and the graph limitation are therefore separable: the first bounds what the QA numbers can mean, and the second holds even in the setting the first excludes. That experiment also quantifies the first reason rather than merely asserting it, since the same reader falls from 84.88 to 43.86 F1 when it must retrieve its own evidence — a larger difference than anything the graph, the ontology or the re-ranker produced, and one that is invisible in any evaluation that keeps the passage in place.

The second is the graph. Its semantic layer is 19.28% conformant, and Section V-D shows both that this understates how compatible the graph is with alternative declarations of its own schema, and that two thirds of its conformance failures involve an endpoint the pipeline never assigned a specific class. Evidence of that quality is consistent with the observed loss from supplying graph content to the reader, which costs 1.49 F1 as a question prefix instead of gaining anything.

The third is the integration. The graph score carries a low weight ($\beta = 0.25$), and Section X-C shows that no larger weight helps: validation F1 falls monotonically as β rises, so the graph score carries no signal this combination rule can exploit. Span re-ranking is in any case a narrow channel: it can only reorder five candidates the reader has already proposed. Only the second and third are fixable, and only the second is addressed by the construction-strategy result.

C. Trustworthiness and utility are different properties

One measurement in this paper is easy to read as inconvenient and is in fact the most useful thing in it. The correlation

between the conformance of the evidence retrieved for a question and the F1 obtained on that question is $r = 0.021$, which is nothing. Questions supported by better evidence are not answered better.

If conformance were a proxy for downstream usefulness, that correlation would be positive. It is not, and two distinct claims follow that the literature routinely merges. In principle, a graph may be schema-conformant and inspectable — its assertions surviving its own schema so that a lawyer could read them — without being useful to a particular reader through a particular integration mechanism. Conversely a graph can move a metric without any of its assertions being inspectable, which is what the worked example in Section XI-G shows: the graph selects the right answer for the wrong reason, promoting a span because a retrieved node string overlapped it.

Conformance measures the first property. It is a precondition for a legal KG that a user is expected to read, and this paper argues it should be reported as a matter of course. On this benchmark, with this integration mechanism and this reader, it does not predict the second, and Section VI-C strengthens that observation with a controlled contrast rather than correlation alone, while still not isolating conformance as the only property that varies: with size and scorer held fixed, almost all of the benefit comes from graph presence and almost none from conformance, which is what it means to say the integration mechanism does not transmit this property. IPROn itself illustrates the second concern rather than the first — not an example of a trustworthy graph, but one whose measurable defects make the distinction visible.

D. Implications for KG-augmented QA

This work draws four lessons that may apply beyond Indian IPR law. The first three concern measurement and are not specific to this benchmark; the fourth is a precondition for the others being worth checking.

First, report baseline strength. The curve shows that the same graph can produce +1.07 F1 or +0.02 F1 depending only on the reader. A single-baseline comparison cannot distinguish a useful KG from a weak baseline.

Second, check that the baseline is free of the intervention. This is the stronger version of the first lesson and the easier one to violate, because a KG-augmented system typically has more than one place to inject graph content, and switching off the obvious one leaves the others running. Here the graph entered both through the re-ranker and through the question, the ablation switches governed only the first, and the resulting baseline was 2.4 to 5.2 F1 below a genuinely graph-free reader. Every reported comparison stayed internally valid and every one of them pointed the wrong way about the graph. The check is mechanical: evaluate the system with all graph inputs removed and confirm that the number equals the baseline being reported.

Third, measure conformance rather than only size. The graph has 17,560 edges and 23 components, which would appear healthy under commonly reported statistics. However, 79.46% of its legal claims violate their declared types in both argument orders. Size and connectivity do not reveal this

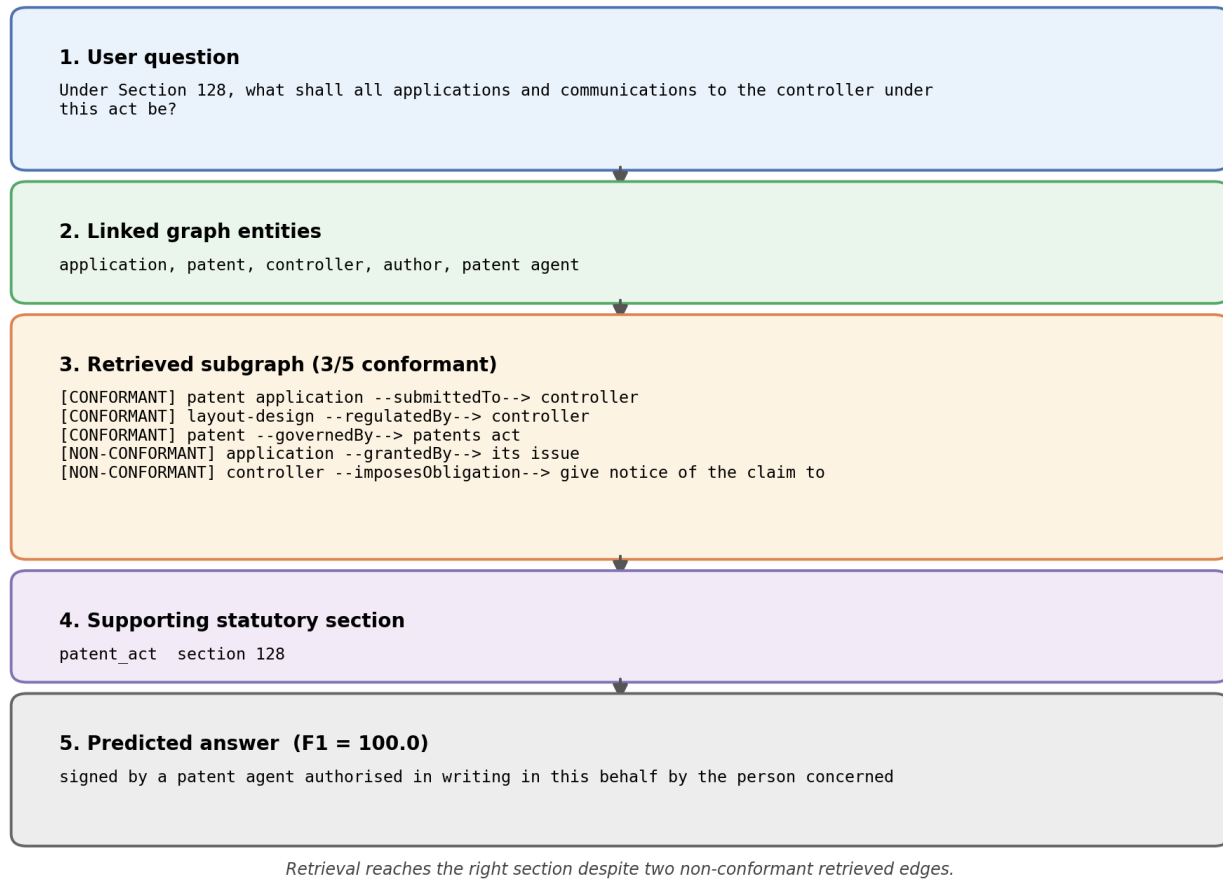


Fig. 4. Retrieval for one test question, shown exactly as the pipeline produced it. This is the cleanest example in the test set: three of five retrieved edges satisfy the ontology-conformance test and two do not. Retrieval reaches the right section despite the two non-conformant edges.

problem. A schema check does, and it runs in seconds. Report it per layer, and report alongside it how much of the failure the schema’s own declarations could account for, as Section V-D does; a conformance figure with neither is hard to interpret and easy to dispute.

Fourth, measure the benchmark before drawing a conclusion from a null result. Two quantities decide whether a QA benchmark can show a knowledge graph working at all: how often the answer is already inside the passage the reader is given, and how often the question already identifies the provision to retrieve. On this benchmark both are high enough to leave essentially no headroom for gains that depend on supplying a missing legal fact (Section III-D), so any KG effect that remains must arise through some other mechanism — span selection, disambiguation, inductive bias — as one does here (Section VI-C). A null on a benchmark of this shape is therefore uninformative about the knowledge-supplying case specifically, not about knowledge graphs in general. Both quantities are one script to compute, and a negative result reported without them cannot be told apart from a benchmark

that never posed the question. Where the second quantity is high, the passage-withheld variant of Section XII is the natural follow-up, and it is worth checking that the masking it requires is actually constructible before promising it.

XIV. THREATS TO VALIDITY

Internal validity. The entity-typing lexicon is rule-based and has not been independently validated. The analysis shows that it makes mistakes, so conformance values that depend on these assignments should be interpreted with caution. The human annotation planned in Section VI-A addresses legal correctness rather than entity-typing accuracy, and therefore does not by itself validate the typer; a separate review packet is released for that purpose. The check in Section VI-A is a partial substitute only: its labels come from language models rather than lawyers, and the two annotators agree at $\kappa = 0.22$, so it supports a direction and not a level. The ablation in Table XXII re-runs inference for each variant rather than reconstructing variants from stored scores, so a variant that is switched off is genuinely switched off. Each variant is still a single run with a fixed seed.

This study is best read as an exploratory forensic audit rather than a confirmatory held-out evaluation. Its findings were reached by investigating anomalies in its own earlier results: a decoder cap, a mislabelled baseline, a duplicated checkpoint, a permissive chain definition, a reconstructed ablation. Each correction was driven by inspecting outputs, and several of those inspections involved the test split. The configuration decisions that remain live — the decoder cap and the score weighting — have since been made on validation data alone (Sections X-D and X-C), but the framing decisions were not made blind, and a reader should treat the effect sizes reported here as estimates from a study that looked at its data rather than as pre-registered tests.

The error attribution in Table XX uses recorded pipeline state rather than manual review. A prediction counted as “reader error despite usable evidence” may therefore contain evidence that a human would judge unusable. The categories are indicative rather than adjudicated.

Construct validity. Conformance measures type correctness rather than legal truth, in both directions. A triple can satisfy its domain and range and still misread the statute, so 19.28% is not a lower bound on error. A triple can also state the law correctly and fail its declared types, which Section V-D shows is common here and the annotation of Section VI-A corroborates, so 19.28% is not an estimate of legal correctness either. It is a measure of whether the graph instantiates the schema published with it, and that is the only claim made for it. Token and character F1 likewise measure overlap rather than legal validity.

The QA construct is narrower still, and now covers two task settings that should not be conflated. In the original extractive setting the gold passage is supplied, so no external fact is ever required (Section III-D) and the experiments measure whether the graph does harm and how easily an apparent benefit can be manufactured. Section XII additionally withholds the passage and requires retrieval, but that remains an easy retrieval condition, because most questions still name the provision they ask about. The benchmark cannot support the stronger setting in which both the passage and the section identifier are absent: under the classification of Table XXIII, no test question survives the removal of its section reference in usable form. None of these results therefore establishes whether a knowledge graph helps legal QA when the relevant provision must be inferred from a naturally authored question, and answering that would require questions written from statutory content rather than from section headings.

Data. The test set is small, with 344 records, and contains redundancy. A total of 162 records fall into 78 groups that share a gold answer, so 84 of the 344 are additional instances of an answer that already appears elsewhere in the split rather than independent items, and one question appears six times. Eleven gold answers contain three words or fewer, which makes exact-match evaluation particularly brittle on those records. These properties limit the resolution of small differences. This is one reason the curve is emphasised over single comparisons and why the validation and test splits are pooled into 752 records for that analysis.

External validity. The QA results cover Indian IPR statutes

only, and cover two task shapes: extractive QA in which the answer is supplied inside the passage, and the passage-withheld statutory retrieval of Section XII. In the second, the original questions usually identify the relevant provision explicitly, so neither setting establishes performance for naturally authored legal questions whose supporting provision is genuinely unknown. The reader-strength curve is measured on that one benchmark and nothing here shows that it holds elsewhere. The validator is more portable, and Section VI-D demonstrates the same code running on two legal KGs built by other authors in another jurisdiction. That demonstration is about the instrument, not about the QA finding, and the external graphs are small and not comparable in kind to those in IPRon. The ontology and extraction patterns remain specific to Indian IPR law.

Reproducibility. Two decoding choices affect absolute scores and neither is a property of the reader. The `[KG: . . .]` question prefix of Section X-E costs 1.49 F1, and reconstructing spans from word-pieces rather than source offsets costs a further 1.96 F1 and 18.60 EM. Comparisons within this paper hold both choices fixed and are therefore unaffected, but absolute levels should not be compared against work that decodes differently, including the companion manuscript [9]. Table XVI reports both baselines for this reason, and its re-rank and net columns answer two different questions. The two free configuration values, the decoder cap and the graph weight, are now selected on validation (Sections X-D and X-C). All primary values reported in this paper are those at the validation-selected cap; every experiment whose conclusion could depend on that choice was also computed at the original test-selected cap and none of them changes, so the residual exposure is to absolute levels rather than to any reported comparison. Local LLM results depend on model versions. For runtime reasons, the embedding metrics use 2,000 validation triples rather than all 11,528, drawn without replacement by `random.sample` under the global seed 42 that the training script sets before evaluation, so the subsample is reproducible. The standard error is about 1.1 points on Hits@1.

XV. LIMITATIONS AND FUTURE WORK

The clearest limitation is that the problem is diagnosed more thoroughly than it is fixed. Prompted extraction is roughly eight times more conformant than the pipeline on matched sentences, and the advantage survives a rebuild at corpus scale (S5 in Table V). The downstream comparison on that larger graph also shows no improvement (Table VI). The pipeline itself remains unfixed. The analysis identifies the relation assigner as a major bottleneck and demonstrates a better alternative, but does not ship a rebuilt production extractor.

The following six directions are concrete next steps.

Replace the relation assigner. Fine-tune InLegalBERT as an 18-way classifier over the IPRon properties and an explicit NONE class. Train it on the 299 annotated triples of Section VI-A, once those labels are human rather than model-generated. The reject class is the critical part: the present assigner has no way to decline. Every candidate above a

0.60 cosine threshold therefore receives a relation, at a mean similarity of 0.699. Report precision and recall for each relation, along with the accuracy and confidence curve that the current scorer cannot produce.

Trace the conformance–coverage curve. Section VI-C evaluates one point on it: rejecting every candidate triple whose arguments violate the declared domain and range yields 304 edges at 100% conformance instead of 1,504 at 19.28%, and costs 0.94 F1 downstream. That is one filter at one threshold. A graded version — accepting a triple when its arguments are compatible with the declared types under a relaxed subsumption, or when a typing model is confident rather than certain — would trace the curve between the two endpoints. The open question is where on that curve a legal user is best served, which is not necessarily either end.

Separate conformance from coverage, and from legal correctness. Section VI-C separates conformance from graph presence and finds the conformance contrast within noise, but two confounds remain that a better experiment would remove. Type enforcement currently buys conformance by discarding four fifths of the edges, so an extractor that reaches high conformance while retaining coverage — by re-extracting the rejected sentences under type constraints rather than dropping them — would separate the two. And conformance is not correctness: with human legal-correctness labels the same design could ask whether accuracy, rather than schema compliance, is what the architecture fails to transmit. Those are different questions, and only the second speaks to whether a legal KG is worth building.

Convert conformance into legal correctness. Repeat the study of Section VI-A with two legally trained human annotators, to estimate what proportion of type-conformant triples are also legally correct. Report Cohen’s κ and adjudicate disagreements. The model-annotated version establishes the direction of the effect but not its level, and the gap between conformance and correctness is exactly what a level would quantify.

Ground generation rather than re-rank spans. The results show that re-ranking is not an effective use of this structure against a strong reader. The typed sections and provenance are better suited to constraining an LLM to answer from retrieved statutory paths. In that setting, a wrong edge is visible to the user rather than silently shifting a span boundary.

Build a benchmark that can pose the retrieval question. Section XII-A shows that the section-blinded condition cannot be constructed from IndicIPR-QA, because its questions were generated from section numbers and titles and 58.72% of the test split consists of nothing but a pointer to a provision. Testing whether a graph helps retrieve an unnamed provision therefore requires questions authored from the content of a section rather than from its heading, with the section reference absent by construction rather than deleted afterwards. A few hundred such questions over the same statutory corpus, doubly annotated and released with their gold sections, would let the retrieval comparison of Section XII-C be repeated in the condition that actually motivates legal knowledge graphs. The per-question classification released with this paper identifies which existing records would have to be rewritten.

XVI. CONCLUSION

This work set out to show that a knowledge graph improves legal question answering. It ended up establishing why its own evidence could not settle the question, and repairing the measurements until that became clear. Three audits carry the result, and they should be read with different scopes.

The graph does not consistently instantiate its own ontology. Only 19.28% of the edges that assert legal facts satisfy their declared argument types, and the residual survives schema completion, endpoint typing and argument reversal. The figure understates how compatible the graph is with alternative declarations of its own schema, since refitting them to the observed data would raise it to 79.96%, but that oracle is fitted to the graph it scores and raises the rate mainly by making the constraint vacuous, so it is not evidence of legal correctness; the dominant defect is that two thirds of the failures involve an endpoint the pipeline never assigned a specific class. Low conformance is nevertheless not inherent to the task, since prompting a local LLM with the same ontology is about eight times more conformant than the pipeline on the matched 300-sentence comparison and 6.8 times more conformant at full-corpus scale.

The evaluation was measuring the wrong difference. The graph reached the answer by two routes, and the baseline carried one of them. Separating them gives 84.88 F1 with no graph input, against 83.39 with the question prefix and 84.45 with re-ranking alone: both routes lower F1 in isolation, and the 0.00 F1 that this system was reported to gain is what results from measuring one route on top of the other. A baseline must be free of the intervention being tested, not merely reported alongside it.

The benchmark could not test the knowledge-supplying and unnamed-provision retrieval claims it was originally used to support. The gold answer lies inside the supplied passage in every one of the 4,462 records and the question names its own statutory section in 94.5% of the test split, so no external fact is ever required and the relevant provision is already named in all but 5.5% of cases. The benefits that motivate a legal KG therefore have almost no room to appear here, which bounds the QA claim to extractive benchmarks of this shape. What remains available is narrower and was measured directly: against a weak reader, re-ranking on a retrieved subgraph is worth +1.75 F1, while making that subgraph more conformant is worth −0.07 F1 with an interval spanning zero, and changes sign at the other decoder cap.

Removing the passage does not change the verdict, and quantifies how far performance moves when supplied evidence is replaced by retrieved evidence. Requiring each system to find the provision among all 1,116 statutory sections, IPROn ranks the correct section first for one test question in 344, against a random floor of one in 1,116. Deleting the graph from the best fused retriever leaves every difference within its bootstrap interval, at the retrieval stage and again after the reader, so no detectable contribution from the graph was observed under the procedure evaluated here. When the supplied passage is replaced by retrieved evidence, the best downstream F1 falls from 84.88 to 43.86. The stronger condition, in which

the section reference is masked as well, could not be built: for 58.72% of test questions the section reference is the whole question, so nothing remains to ask once it is removed. That measurement is itself a contribution, because it establishes that this benchmark cannot pose the retrieval task the field assumes it poses.

The general lesson is the order of operations. Graph quality, baseline purity and benchmark headroom are all cheap to measure and all decide what a QA comparison can mean, so they belong before the comparison rather than in its limitations. What generalises here is accordingly the instruments rather than the graph. The conformance audit is inexpensive, layered, and transfers unchanged to two legal KGs published by other authors, where it finds an ontology that no compliant parser will load. The baseline check is a single run of the system with every graph input removed. The headroom check is two counts over the benchmark, and the passage-withheld variant that follows it is a retrieval run over the corpus the benchmark was built from. None is novel as reasoning; all answer questions that ontology-guided KG papers rarely ask. IPRon, the validator, the audit scripts and the reports behind every table are released.

DATA, CODE AND ETHICS

The IndicIPR-QA dataset is available at <https://huggingface.co/datasets/niloycste68/IndicIPRExtractiveQA>. The pipeline is organised in five phases. The code, the audit scripts, the generated graphs, the configuration and seeds, and the JSON report behind every table are provided with this submission as a supplementary archive, and will be placed in a public repository with a citable commit hash on publication. Statements elsewhere in this paper that an artifact is released refer to that archive rather than to a currently public repository. The system is intended for legal information access and research, not for legal advice. Outputs may omit statutory conditions, exceptions, or provisos. Any downstream use requires expert review.

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